CHAPTER 1

General Issues in Understanding and Treating Psychopathology

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The practice of any mental health profession (clinical and counseling psychology, psychiatry, social work) depends heavily on the ability to recognize and conceptualize various forms of mental disorder. Therapists, irrespective of their professional background and theoretical orientation, must be aware of the varied manifestations of psychopathology. They also have to understand many forms of vulnerability that set the stage for the development and maintenance of commonly occurring mental disorders that eventually touch all of our lives, either directly or indirectly. Mental disorders are a leading cause of disability around the world, and within the United States, neuropsychiatric disorders are the leading cause of disease-related disability and risk for early mortality (Murray et al., 2013; Whiteford, Ferrari, Degenhardt, Feigin, & Vos, 2015).

The individual chapters in this book provide truly unique reviews of the professional literature concerned with specific forms of psychopathology. Each is written by a team of scholars that includes recognized experts in the treatment of the condition, as well as leaders in the scientific study of the causes of the disorder. They have worked together to produce a creative synthesis describing the nature of the disorder (e.g., symptoms, course, epidemiology, and etiology) and the treatment implications that follow from this knowledge (assessment, case formulation, and/or principles of change). These chapters represent classic examples of the thoughtful integration of science and practice: Clinical experience raises important research questions, and evidence from scientific studies leads to the development and evaluation of improved treatment procedures. In this opening chapter, we discuss several basic issues that lay the foundation for subsequent chapters discussing specific types of psychopathology.

DEFINING MENTAL DISORDERS

Any systematic review of what is known about psychopathology first requires attention to how the construct of "mental disorder" is defined. Should some (if not all) mental disorders be conceptualized (completely or in part) as diseases, character flaws, deficits, or problems of living? How are mental disorders distinguished from behaviors that are simply idiosyncratic, eccentric, or out of favor within a particular culture? These questions determine, sometimes explicitly but most often implicitly, how a mental health professional responds to a particular client or patient, including issues such as whether the person should receive treatment, and what kind of treatment should be provided. Many attempts have been made to define abnormal behavior, but none is entirely satisfactory. No one has provided a universal definition that can account for all situations in which the concept of mental disorder is invoked (Pilgrim, 2005; Zachar & Kendler, 2007).

Arguably, one of the most influential and widely invoked definitions of "mental disorder" was proposed by Wakefield (1992, 2010). According to his argument, a behavioral condition should be considered a mental disorder if, and only if, it meets two criteria:

1. The condition results from the inability of some internal mechanism (mental or physical) to perform its natural function. In other words, something inside the person is not working properly. Examples of such mechanisms include those that regulate levels of emotion, and those that distinguish between real auditory sensations and imagined ones.
2. The condition causes some harm to the person as judged by the standards of the person's culture. These negative consequences are measured in terms of the person's own subjective distress or difficulty performing expected social or occupational roles.

Using Wakefield's terms, mental disorders are defined as "harmful dysfunctions." One element ("dysfunction") of this definition is an attempt to incorporate (as much as it is possible) an objective evaluation of behavior. Wakefield argues that internal processes (e.g., cognition and perception) have a natural function, and that this function is to allow an individual to perceive the world in ways that are shared with other people and to engage in rational thought, problem solving, and adaptive behavior. The dysfunctions in mental disorders are assumed to be the product of disruptions of thought, feeling, communication, perception, and motivation.

This view of mental disorder also recognizes that all types of dysfunction do not lead to a disorder. Only dysfunctions that result in significant harm to the person are considered to be disorders. There are, for example, many types of physical dysfunction, such as albinism, reversal of heart position, and fused toes, that clearly represent a significant departure from the way that some biological process ordinarily functions. These conditions are not considered to be disorders, however, because they are not necessarily harmful to the person.

By definition, mental disorders are harmful to the person's adjustment. They typically affect family relationships, as well as success in educational and occupational activities. There are, of course, other types of harm associated with mental

disorders. These include subjective distress, such as high levels of anxiety or depression, as well as more tangible outcomes, such as suicide.

The definition of abnormal behavior presented in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5; American Psychiatric Association, 2013), which in the United States is considered a standard diagnostic system, incorporates many of the factors we have already discussed. According to this definition, a "mental disorder" is "a syndrome characterized by clinically significant disturbance in an individual's cognition, emotion regulation, or behavior that reflects a dysfunction in the psychological, biological, or developmental processes underlying mental functioning" (p. 20). The manual also notes that mental disorders are associated with subjective distress or impairment in social functioning. DSM-5 also excludes several conditions from consideration as mental disorders. These include (1) an expectable or culturally sanctioned response to a particular event (e.g., death of a loved one); (2) socially deviant behavior (e.g., the actions of political, religious, or sexual minorities); and (3) conflicts that are between the individual and society (e.g., voluntary efforts to express individuality).

The DSM definition places primary emphasis on the consequences of certain behavioral syndromes. Accordingly, mental disorders are defined by clusters of persistent, maladaptive behaviors that are associated with personal distress, such as anxiety or depression, or with impairment in social functioning, such as job performance or personal relationships. This definition, therefore, recognizes the concept of dysfunction, and it spells out ways in which the harmful consequences of a disorder might be identified.

The practical boundaries of abnormal behavior are defined by the list of disorders included in the DSM. Therefore, the manual provides a simplistic, though practical, explanation of how someone's behavior might be considered pathological: It would be considered abnormal if the person's experiences fit the description of one of the forms of mental disorder listed in the diagnostic manual. Conceptually, however, it could be argued that a valid and adequate description of a mental disorder, such as major depression, cannot be restricted to its symptomatic picture. In order to define and understand major depression, it is necessary to consider a description of other clinical features (e.g., marital and health problems with which it is frequently associated), as well as a number of additional issues, such as its typical course (onset, duration, recurrence, relapse), prevalence (across gender and culture), patterns of comorbidity, and vulnerability factors.

In an effort to move beyond the DSM-5 definition of mental disorder, many researchers have moved toward empirical approaches of mapping dimensions of dysfunction and their interrelatedness as opposed to relying on categorical diagnoses. The hierarchical taxonomy of psychopathology (HiTOP) is a dimensional model of psychopathology that includes most mental disorders (e.g., mood, anxiety, substance use, psychotic, and personality disorders) (Kotov et al., 2017). At the highest level is a general factor of psychopathology (*p* factor), which has been replicated across several samples in many contexts (Caspi et al., 2014; Smith, Atkinson, Davis, Riley, & Oltmanns, 2020). The *p* factor is shared across all dimensions of psychopathology and represents an index of nonspecific impairment. Beneath the *p* factor are three broad domains of internalizing disorders, externalizing disorders, and thought disorder. Many argue that the structure of general personality is fundamental to

understanding the HiTOP dimensional model of psychopathology, and that in order to understand and define mental disorders, one must consider personality as a core component to the manifestation of psychopathology (Widiger et al., 2019). Another definition of psychopathology that has recently received attention is "network theory," an approach suggesting that mental disorders arise from the causal interaction between symptoms within a given network (Borsboom, 2017).

Mental disorders are actually best conceived as "hypothetical constructs" (Morey, 1991; Neale & Oltmanns, 1980), which simply are abstract, explanatory devices. In the case of behavioral disorders, a hypothetical construct (i.e., a particular mental disorder) is an internal event whose existence is inferred on the basis of observable behaviors and the context in which they occur. The construct itself cannot be directly observed, but it is tied to overt referents that can be observed. For example, we cannot measure major depression directly, but we can see that the person no longer enjoys formerly pleasurable activities, has withdrawn from social contacts, is sleeping more than usual, has lost their appetite, and frequently talks about feeling worthless. Beyond these specific symptoms, the utility of a hypothetical construct ultimately depends on the extent to which it enters into relationships with other constructs and observable events (interrelated dimensions of human functioning). With this approach, a mental disorder is therefore defined by more than simply the diagnostic criteria identified in a manual such as DSM-5.

Clinically, this approach to defining a disorder (e.g., major depression) does not imply that clinicians should disregard "official" diagnostic criteria. Rather, it suggests that in order to conduct more comprehensive assessments and case formulations of their clients, as well as to identify a more complete list of potential targets for intervention, therapists should complement their diagnostic (DSM-based) evaluation with a careful consideration of nonsymptomatic factors that have been empirically associated with this disorder. One of our goals in this book is to provide a review of research on the factors that appear to contribute to the nature of each of the several specific forms of psychopathology.

In the chapters to come, authors focus on describing mental disorders, the impact of those disorders on people who suffer from them, the factors that provoke and maintain these disorders, and relevant evidence-based strategies for clinical interventions to alleviate them and/or help patients manage them. These chapters do not delineate specific criteria from the official diagnostic manuals that are widely in use (the DSM and the *International Classification of Diseases* [ICD]). If you are interested in those criteria, we encourage you to review them in the manuals themselves: DSM-5 (American Psychiatric Association, 2013) and ICD-11 (World Health Organization, 2019).

EPIDEMIOLOGY

Basic understanding of mental disorders requires data about the frequency with which these disorders occur. Moreover, important clinical, organizational, and social decisions can be based on epidemiological information, such as whether the frequency of a disorder has increased or decreased during a particular period, whether it is more common in one geographic area than in another, and whether

certain types of people—based on factors such as gender, ethnicity, and socioeconomic status—are at greater risk than other types for the development of the disorder. Health administrators often use such information to make decisions about the allocation of resources for professional training programs, treatment facilities, and research projects. As described in this book, answers to such questions have also provided insights and generated further research on possible risk factors for specific disorders.

How prevalent are the various forms of abnormal behavior? One important dataset regarding this question comes from a large-scale study known as the National Comorbidity Survey Replication (NCS-R) conducted between 2001 and 2003 (Kessler et al., 2005; Kessler, Merikangas, & Wang, 2007). We use this study to illustrate an overall impression regarding the prevalence of mental disorders because it collected evidence of many major forms of mental disorders simultaneously in one large group of individuals. Specifically, members of the NCS-R research team interviewed a nationally representative sample of approximately 9,000 people living in the continental United States. Questions were asked pertaining to several (but not all) of the major disorders listed in DSM-IV (American Psychiatric Association, 1994). The NCS-R found that 46% of the people they interviewed received at least one *lifetime* diagnosis (meaning that they met the criteria for one of the disorders assessed at some point during their lives), with first onset of symptoms usually occurring during childhood or adolescence. This proportion of the population is much higher than many people expect, and it underscores the point we made at the beginning of this chapter: All of us can expect to encounter the challenges of a mental disorder—either for ourselves or for someone we love—at some point during our lives.

Figure 1.1 illustrates some results from this study using 12-month prevalence rates—the percentage of people who had experienced each disorder during the most recent year. The most prevalent specific types of disorder were major depression (17%) and alcohol abuse (13%). Various kinds of anxiety disorders were also relatively common. Substantially lower prevalence rates were found for obsessive–compulsive disorder and bipolar disorder, which each affect approximately 2% of the population. These lifetime prevalence rates are consistent with data reported by other, subsequent studies regarding the prevalence of mental disorders. More specific and recent evidence regarding specific forms of mental disorder are presented in each of the later chapters in this book.

Although many mental disorders are quite common, they are not always seriously debilitating. The NCS-R investigators assigned each case a score with regard to severity, based on the magnitude and number of symptoms, as well as the level of occupational and social impairment that the person experienced. Averaged across all of the disorders diagnosed in the past 12 months, 40% of cases were rated as *mild*, 37% as *moderate*, and only 22% as *severe*. Mood disorders were most likely to be rated as severe (45%), while anxiety disorders were less likely to be rated as severe (23%). These findings have important clinical (and social) implications, as they suggest that not all individuals meeting criteria of a diagnosis necessarily need immediate treatment. At the same time, it is worth noting that interventions (of varied length and intensity) might be beneficial for some individuals suffering from low-level distress and impairment, as subclinical levels of symptoms can, for some disorders, signal the eventual emergence of acute and severe symptoms.

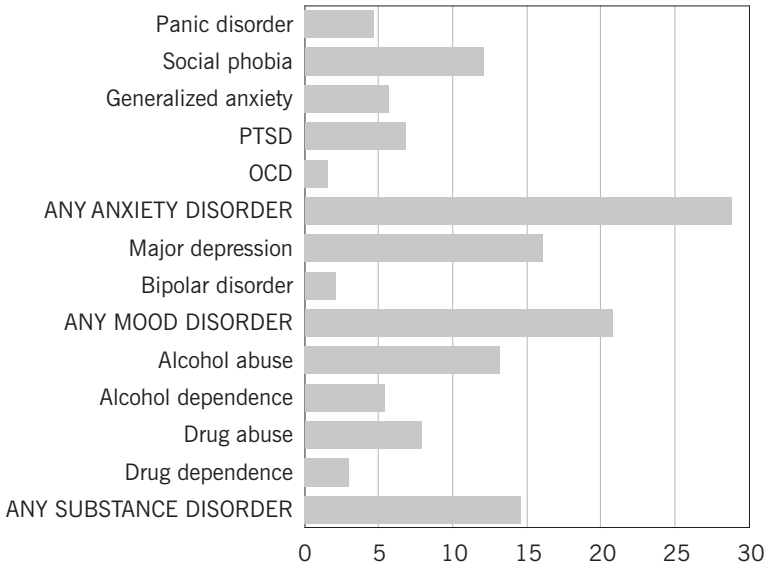

FIGURE 1.1. Twelve-month prevalence rates for various mental disorders (NCS-R data). Data from Kessler, Merikangas, and Wang (2007). PTSD, posttraumatic stress disorder; OCD, obsessive–compulsive disorder. The NCS-R used DSM-IV definitions of mental disorders. Thus, PTSD and OCD were still considered forms of anxiety disorder, and the distinction between substance dependence and substance abuse was still recognized.

Epidemiological studies such as the NCS-R have consistently found gender differences for many types of mental disorder: Major depression, anxiety disorders, trauma and stress-related disorders, and eating disorders are more common among women; alcoholism and antisocial personality disorders are more common among men. Patterns of this sort are highlighted throughout the chapters in this book as clues for possible causal mechanisms in some of these disorders.

Epidemiologists also measure the extent of mental disorders' impact on people's lives. The Global Burden of Disease Study 2010 included a systematic review of all epidemiological data on diseases, injuries, and related risk factors across 187 countries from 1990 to 2010. Whiteford and colleagues (2015) examined the specific global burden of mental, neurological, and substance use disorders from that sample in 2010 and found that these disorders accounted for 10.4% of global disability-adjusted life years and 28.5% of global years lived with disability; mental disorders represented the largest proportion of disability-adjusted life years (56.7%) among these types of disorders. Within the United States, mental and behavioral disorders are the largest contributor to years lived with disability, and in particular, major depression, anxiety disorders, and substance use disorders are among the top 15 contributors to chronic disability and early mortality (Murray et al., 2013). The burden of mental disorders has greatly increased in the past 10 years and will likely continue to increase by the year 2030. These surprising results indicate strongly that mental disorders are one of the world's greatest health challenges.

CULTURE AND PSYCHOPATHOLOGY

As evidence regarding the global burden of disease clearly documents, mental disorders affect people all over the world. That does not mean, however, that the symptoms of psychopathology and the expression of emotional distress take the same form in all cultures. Epidemiological studies comparing the frequency of mental disorders in different cultures suggest that some disorders, such as schizophrenia, show important consistencies in cross-cultural comparisons. They are found in virtually every culture that social scientists have studied. Other disorders, such as bulimia nervosa, are more specifically associated with cultural factors, as revealed by comparisons of prevalence in different parts of the world and changes in prevalence over several generations. Almost 90% of patients with bulimia are women. Within the United States, the incidence of bulimia is much higher among university women than among working women, and it is more common among younger women than among older women (Udo & Grilo, 2018). The prevalence of bulimia is much higher in Western nations than in other parts of the world. Furthermore, the number of cases increased dramatically during the latter part of the 20th century (Keel & Klump, 2003). As discussed later in this book, these patterns suggest that holding particular sets of values related to eating and to women's appearance is an important ingredient in establishing risk for development of an eating disorder.

The strength and nature of the relationship between culture and psychopathology varies from one disorder to the next. Several general conclusions can be drawn from cross-cultural studies of psychopathology (Draguns & Tanaka-Matsumi, 2003), including the following points:

- All mental disorders are shaped, to some extent, by cultural factors.
- No mental disorders are entirely due to cultural or social factors.
- Psychotic disorders are less influenced by culture than are nonpsychotic disorders.
- The symptoms of certain disorders are more likely to vary across cultures than are the disorders themselves.

Clinicians must consider the influence of cultural factors in both the expression and recognition of symptoms of mental disorders. People express extreme emotions in ways that are shaped by the traditions of their families and other social groups to which they belong. Intense, public displays of anger or grief might be expected in one culture but be considered signs of disturbance in another. Interpretations of emotional distress and other symptoms of disorder are influenced by the explanations that a person's culture assigns to such experiences. Religious beliefs, social roles, and sexual identities all play an important part in constructing meanings that are assigned to these phenomena (Hwang, Myers, Abe-Kim, & Ting, 2008). The most obvious clinical implication one can derive from these important issues is that the accuracy and utility of a clinical diagnosis depend on more than a simple count of the symptoms that appear to be present. They also hinge on the clinician's ability to consider the cultural context in which the problem appeared. This is a particularly challenging task when the clinician and the person with the problem do not share the same cultural background.

Clinicians can also become sensitized to the importance of cultural issues by considering cultural concepts of distress, patterns of erratic or unusual thinking and behavior that have been identified in diverse societies around the world and do not fit easily into the other diagnostic categories. A list of these syndromes is included in the appendix of DSM-5. They have also been called "culture-bound syndromes," because they are considered to be unique to particular societies, particularly in non-Western or developing countries. Their appearance is easily recognized and understood to be a form of abnormal behavior by members of certain cultures, but they do not conform to typical patterns of mental disorder seen widely across areas in the United States or Europe. Cultural concepts of distress have also been called "idioms of distress." In other words, they represent a manner of expressing negative emotion that is unique to a particular culture and cannot be easily translated or understood in terms of its individual parts.

One syndrome of this type is a phenomenon known as *ataques de nervios,* which has been observed most extensively among people from Puerto Rico and other Caribbean countries (Lewis-Fernández et al., 2002; San Miguel et al., 2006). Descriptions of this experience include four dimensions, in which the essential theme is loss of control—an inability to interrupt the dramatic sequence of emotion and behavior. These dimensions include emotional expressions (an explosion of screaming and crying, coupled with overwhelming feelings of anxiety, depression, and anger), bodily sensations (including trembling, heart palpitations, weakness, fatigue, headache, and convulsions), actions and behaviors (dramatic, forceful gestures that include aggression toward others, suicidal thoughts or gestures, and trouble eating or sleeping), and alterations in consciousness (marked feelings of "not being one's usual self," accompanied by fainting, loss of consciousness, dizziness, and feelings of being outside of one's body).

Ataques are typically provoked by situations that disrupt or threaten the person's social world, especially the family. Many *ataques* occur shortly after the person learns unexpectedly that a close family member has died. Others result from an imminent divorce or occur after a serious conflict with a child. Women are primarily responsible for maintaining the integrity of the family in this culture, and they are also more likely than men to experience *ataques de nervios*. Puerto Rican women from poor and working-class families define themselves largely in terms of their success in building and maintaining a cohesive family life. When this social role is threatened, an *ataque* may result. This response to threat or conflict—an outburst of powerful, uncontrolled negative emotion—expresses suffering, while simultaneously providing a means for coping with the threat. It serves to signal the woman's distress to important other people and to rally needed sources of social support.

What is the relation between cultural concepts of distress and the formal categories listed in the DSM or the ICD? The answer is unclear and also varies from one syndrome to the next. Are they similar problems that are simply given different names in other cultures? Probably not, at least not in most instances (Guarnaccia & Pincay, 2008). In some cases, people who exhibit behavior that would fit the definition of a culture-bound syndrome would also qualify for another diagnosis, if diagnosed by a clinician trained in the use of the diagnostic manual (Tolin, Robison, Gaztambide, Horowitz, & Blank, 2007). But not everyone who displays the culture-bound

syndrome meets criteria for another mental disorder, and of those who do, not all would receive the same diagnosis.

The DSM's glossary on cultural concepts of distress has been praised as a significant advance toward integrating cultural considerations into the classification system (Trinh, Son, & Chen, 2019). It has also been criticized for its ambiguity. The most difficult conceptual issue involves the boundary between culture-bound syndromes and categories found elsewhere in the diagnostic manual. Some critics have noted that if culturally unique disorders must be listed separately from other, "mainstream" conditions, then certain disorders now listed in the main body of the manual—especially eating disorders, such as bulimia—should actually be listed as culture-bound syndromes. Like *ataques de nervios,* bulimia nervosa is a condition that is found primarily among a limited number of cultures (Keel & Klump, 2003). "Dissociative amnesia"—the inability to recall important personal information regarding a traumatic event—also resembles culture-bound syndromes because it appears to be experienced only by people living in modern, developed cultures (Pope, Poliakoff, Parker, Boynes, & Hudson, 2007).

Though it is imperfect, DSM-5's list of cultural concepts of distress raises important conceptual and clinical implications. The fact that some diagnostic categories that are familiar to most mental health professionals working in Western or developed countries are unique to their cultural environment helps to challenge the frequently held assumption that culture shapes only conditions that appear to be exotic in faraway lands. At a more direct, clinical level, the glossary serves to make clinicians more aware of the extent to which their own views of what is normal and abnormal have been shaped by the values and experiences of their own culture (Mezzich, Berganza, & Ruiperez, 2001).

DSM-5 also now includes the Cultural Formulation Interview (CFI), a supplemental module that serves as a semistructured assessment tool to assist providers in clarifying the contribution of culture to clinical symptoms at any point during clinical care (Lewis-Fernández et al., 2014). The core interview, the first and main component of the CFI, includes 16 open-ended questions related to a cultural definition of the psychological problem, cultural perceptions related to the problem (e.g., with regard to cause, context, and support), and cultural factors affecting coping or help-seeking behaviors. Other components of the CFI include an informant report and other supplemental modules that can be used as needed. The CFI can be used with any client, regardless of whether a clinician has questions about how their culture may impact the presentation of clinical symptoms, and the CFI serves as a valuable tool to ensure that relevant and important cultural factors are not missed during case conceptualization and throughout treatment.

CLASSIFICATION OF PSYCHOPATHOLOGY

One might argue that efforts to classify mental disorders are as important and have raised as much controversy as efforts to define them. Despite many tribulations and debates associated with the classification of psychopathology, this process has served several purposes. A classification system helps clinicians to (1) identify (as accurately

and comprehensively as possible) their clients' problems and (2) formulate a plan of intervention that is most likely to be effective for these difficulties. A classification system also serves an important role in the search for new knowledge. The history of medicine is filled with examples of problems that were recognized long before they could be treated successfully. The classification of a specific set of symptoms has often laid the foundation for research that eventually identified a cure or a way of preventing the disorder.

Modern classification systems in psychiatry were introduced shortly after World War II. During the 1950s and 1960s, psychiatric classification was widely criticized. One major criticism focused on the lack of consistency in diagnostic decisions (Nathan & Langenbucher, 2003). Independent clinicians frequently disagreed with one another about the use of diagnostic categories. Objections were also raised from philosophical, sociological, and political points of view. For example, some critics charged that diagnostic categories in psychiatry would be more appropriately viewed as "problems in living" than as medical disorders (Szasz, 1963). Others were concerned about the negative impact of using diagnostic labels. In other words, once a psychiatric diagnosis is assigned, the person so labeled might experience discrimination of various kinds, and also find it more difficult to establish and maintain relationships with other people. These are all serious problems that continue to be the topic of important, ongoing discussions involving mental health professionals, as well as patients and their families. Debates regarding these issues did fuel important improvements in the diagnosis of mental disorders, including emphasis on the use of detailed criterion sets for each disorder.

Currently, two diagnostic systems for mental disorders are widely recognized. We have already mentioned DSM-5, which is published by the American Psychiatric Association. The other—the ICD—is published by the World Health Organization. Both systems were first developed shortly after World War II and have been revised several times. The World Health Organization's manual is in its 11th edition and is therefore known as ICD-11. Deliberate attempts are made to coordinate the production of DSM and ICD manuals. Most of the categories listed in the manuals are identical, and the criteria for specific disorders are usually similar. More than 200 specific diagnostic categories are described in the current edition of DSM. Importantly, under both systems (DSM and ICD), a person can be assigned more than one diagnosis if they meet criteria for more than one disorder.

EVALUATION OF PSYCHIATRIC CLASSIFICATION

The most important issue in the evaluation of a diagnostic category is whether it is *useful* (Kendell & Jablensky, 2003). By knowing that a person fits into a particular group or class, do we learn anything meaningful about that person? For example, is the person who fits the diagnostic criteria for schizophrenia likely to improve when they are given antipsychotic medication? Or is that person likely to have a less satisfactory level of social adjustment in five years than a person who meets diagnostic criteria for bipolar mood disorder? Does the diagnosis tell us anything about the factors or circumstances that might have contributed to the onset of this problem? These questions are concerned with the validity of the diagnostic category.

Validity is, in a sense, a reflection of the success that has been achieved in understanding the nature of a disorder. Have important facts been discovered? Systematic studies aimed at establishing the validity of a disorder may proceed in a sequence of phases (Robins & Guze, 1989). After a clinical description has been established, diagnostic categories are refined and validated through a process of scientific exploration. Unfortunately, relatively few of the disorders listed in the official diagnostic manual are supported by an extensive set of research evidence supporting all possible aspects of validity.

It may be helpful to think of different forms of validity in terms of their relationship in time with the appearance of symptoms of the disorder. *Etiological validity* is concerned with factors that contribute to the onset of the disorder. These are things that have happened in the past. Was the disorder regularly triggered by a specific set of events or circumstances? Did it run in families? The ultimate question with regard to etiological validity is whether there are any specific causal factors that are regularly, and perhaps uniquely, associated with this disorder. If we know that a person exhibits the symptoms of the disorder, do we in turn learn anything about the circumstances that originally led to the onset of the problem?

Concurrent validity is concerned with the present time and with correlations between the disorder and other symptoms, circumstances, and test procedures. Is the disorder currently associated with any other types of behavior, such as performance on psychological tests? Do precise measures of biological variables, such as brain structure and function, distinguish reliably between people who have the disorder and those who do not? Clinical studies aimed at developing a more precise description of a disorder also fall into this type of validity.

Predictive validity is concerned with the future and with the stability of the problem over time. Will it be persistent? If it is short-lived, how long will an episode last? Will the disorder have a predictable outcome? Do people with this problem typically improve if they are given a specific type of medication or a particular form of psychotherapy?

The overall utility of a diagnostic category depends on the body of evidence that accumulates as scientists seek answers to questions raised by these multiple forms of validity (*reliability*, or consistency of the diagnostic decisions, is also crucial, as it precedes validity). At this point in time, some disorders included in the diagnostic manual are based on a much more extensive foundation of evidence than others. Thus, more research needs to confirm and enhance the validity (thus, the utility) of our current diagnostic categories. Such research should be conducted in parallel with other efforts (conceptual and empirical) to address serious criticism directed toward the diagnostic manual.

LIMITATIONS OF AND POSSIBLE IMPROVEMENTS TO THE DSM

Developing each edition of the DSM involves a tremendous amount of work (with regard to both theory and research), and each new edition represents an improvement over previous ones (see Frances & Widiger, 2012). Nevertheless, several enduring issues remain to be resolved. One fundamental question that applies to every disorder involves the boundary between normal and abnormal behavior. DSM-5

is based on a categorical approach to classification, but most of the symptoms that define the disorders are actually dimensional in nature. Depressed mood, for example, can vary continuously, from the complete absence of depression to moderate levels of depression, on up to severe levels of depression. The same issue applies to symptoms of anxiety disorders, eating disorders, and substance use disorders. These are all continuously distributed phenomena, and there is not a bright line that divides people with problems from those who do not have problems. DSM-5 does include a greater focus on dimensional conceptualizations of certain disorders (e.g., including a spectrum for schizophrenia and for substance use disorders, the alternative model of personality disorder in Section III) in comparison to DSM-IV. Nevertheless, the DSM remains a categorical classification of separate disorders.

The absence of a specific definition of social impairment is another conceptual and clinical issue that has plagued the current diagnostic manual. Most disorders in the DSM include the requirement that a particular set of symptoms causes "clinically significant distress or impairment in social or occupational functioning." However, no specific measurement procedures are provided to make this determination. Mental health professionals must rely on their own subjective judgment of a person's symptoms to decide how distressed or how impaired a person must be in order to qualify for a diagnosis. There is an important need for more specific definitions of these concepts, and better measurement tools are needed for their assessment.

In addition, criticisms of the current classification system have emphasized broad conceptual issues. Some clinicians and investigators have argued that the syndromes defined in the DSM do not represent the most useful ways to think about psychological problems, in terms of either planning current treatments or designing programs of research. For example, it might be better to focus on more homogeneous dimensions of dysfunction, such as anxiety or angry hostility, rather than on syndromes (groups of symptoms) (Smith & Combs, 2010). Clinically and empirically, this raises questions such as the following: Should we design treatments for people who exhibit distorted, negative ways of thinking about themselves, regardless of whether their symptoms happen to involve a mixture of depression, anxiety, or some other pattern of negative emotion or interpersonal conflict? Indeed, in recent years there has been a greater move toward the development and implementation of transdiagnostic treatments for mental disorders (e.g., the unified protocol for transdiagnostic treatment of emotional disorders; Barlow et al., 2017).

No definite answer is available at this point in time. It would certainly be premature to cut off consideration of these alternatives just because they address problems in a way that deviates from the official diagnostic manual. In our current state of uncertainty, diversity of opinion should be encouraged, particularly if it is grounded in cautious skepticism and supported by rigorous scientific inquiry.

From an empirical and clinical point of view, the DSM is hampered by a number of problems that suggest it does not classify clinical problems into syndromes in the simplest and most beneficial way (Helzer, Kraemer, & Krueger, 2006). One of the thorniest issues involves "comorbidity," which is defined as the simultaneous appearance of two or more disorders in the same person. Comorbidity rates are very high for mental disorders as defined in the DSM system (Eaton, South, & Krueger, 2010). For example, in the NCS, among those people who qualified for at least one diagnosis at some point during their lifetime, 56% met the criteria for two or more

disorders. A small subgroup, 14% of the sample, actually met the diagnostic criteria for three or more lifetime disorders. That group of people accounted for almost 90% of the *severe* disorders in the study.

There are several ways to interpret comorbidity (Krueger, 2002). Some people may independently develop two separate conditions. In other cases, the presence of one disorder may lead to the onset of another. Unsuccessful attempts to struggle with prolonged alcohol dependence, for example, might lead a person to become depressed. Neither of these alternatives creates conceptual problems for the DSM; it makes sense that some people have more than one disorder and that one disorder can lead to another. Unfortunately, the high rates of comorbidity that have been observed empirically suggest that these explanations account for a small proportion of overlap between categories. In a large proportion of cases, the overlapping manifestations of more than one disorder do not seem to reflect clearly distinct diagnostic constructs.

Major clinical problems associated with comorbidity can arise when a person with a mixed pattern of symptoms, usually of a severe nature, simultaneously meets the criteria for more than one disorder. Consider, for example, a client who was treated by one of us (T. F. O.). This man experienced a large number of diffuse problems associated with anxiety, depression, and interpersonal difficulties. According to the DSM system, he would have met the criteria for major depressive disorder, generalized anxiety disorder, and obsessive–compulsive disorder, as well as three types of personality disorder. It might be said, therefore, that he suffered from at least six types of mental disorder. But is that really helpful? Is it the best way to think about his problems? Would it be more accurate to say that he had a complicated set of interrelated problems that were associated with worrying, rumination, and the regulation of high levels of negative emotion, and that these problems constituted one complex and severe type of disorder?

The comorbidity issue is related to another limitation of the DSM: the failure to make better use of information regarding the course of mental disorders over time. More than 100 years ago, when schizophrenia and bipolar mood disorder were originally described, the distinction between them was based heavily on observations regarding their long-term course. Unfortunately, most disorders listed in the DSM are defined largely in terms of snapshots of symptoms at particular points in time. Diagnostic decisions are seldom based on a comprehensive analysis of the way that a person's problems evolve over time. If someone meets the criteria for more than one disorder, does it matter which one came first? Is there a predictable pattern in which certain disorders follow the onset of others? What is the nature of the connection between childhood disorders and adult problems? Our knowledge of mental disorders would be greatly enriched if greater emphasis were placed on questions regarding lifespan development (Buka & Gilman, 2002; Oltmanns & Balsis, 2011).

Progress on understanding some of the traditional diagnostic categories in the DSM has been quite slow over the past 50 years, and it has led many to wonder whether there are better ways to approach classification and thus treatment. A number of alternative models of psychopathology have been developed as a counter to the "top-down," categorically based DSM classification system. HiTOP, as described earlier, was developed to understand the natural organization of psychopathology (Kotov et al., 2017) and emerged with a hierarchical dimensional structure of psychopathology, including a single general *p* factor and three broad domains under the

p factor (internalizing, externalizing, and thought disorder) (Smith et al., 2020). The Research Domain Criteria (RDoC; Insel et al., 2010) framework was created by the National Institute of Mental Health to encourage the development of a dimensional classification system of psychological processes that link with specific neural bases and are related to various aspects of psychopathology. A main goal of RDoC is to understand biological processes relevant to psychopathology, and many researchers are now taking this approach to studying psychopathology. Whether a "bottom-up" approach will yield better results in understanding clinical problems is a question that is likely to remain at the forefront of the mental health field in the decade ahead.

The classification issues identified earlier are being considered by the experts who develop each new edition of the DSM and the ICD. Of course, all of them will not be solved immediately. Attempts to provide solutions to these problems and limitations ensure that the classification system will continue to be revised. As before, these changes will be driven by the interaction of clinical experience and empirical evidence. Students, clinicians, and research investigators should all remain skeptical when using this classification system and its successors. At the same time, the complexity of psychopathology and its treatment should encourage all scholars and providers in the field of mental health to pay close attention to our current knowledge, and to derive from it new and creative ways to think about and intervene with mental disorders. The following chapters in this book have been written with this goal in mind.

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CHAPTER 2

Depression

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It is estimated that 16% of the general population experiences clinically significant depression in a given 12-month period (Kessler, Tat Chiu, Demler, & Walters, 2005). In addition to the impact on affected individuals and their families, depressive disorders place a burden of almost $50 billion per year on the American economy, accounting for over 20% of costs for all mental illness (Stewart, Ricci, Chee, Hahn, & Morganstein, 2003). Compared to other physical and mental disorders, depression is the leading cause of disability worldwide according to the World Health Organization (James et al., 2018). Given the substantial personal and societal costs of this disorder, efforts to identify vulnerability factors and effective interventions for depression are particularly important.

Our goal in this chapter is to present basic psychopathology research in depression and to discuss the treatment implications of these findings. Though all depressive disorders share similar features (Gotlib & LeMoult, 2014), there are some important differences between diagnostic categories that are not covered in this chapter. Instead, we focus here on depressive symptoms in general and major depressive disorder (MDD) in particular. We first describe the phenomenology of depression, including its associated clinical features and course. We then discuss the etiology of depression and focus in particular on novel empirical findings. After each section, we address the clinical implications of the findings. This is a timely and important task because, even though effective depression treatments exist, these interventions have undergone few changes in the past decades, and rates of recurrence of depression are still high even after successful treatment.

Cognitive-behavioral therapy (CBT), for example, is currently one of the most empirically supported and effective treatments for this emotional disorder. It is fascinating to see, however, that CBT for depression has undergone few changes in the last 50 years. This is surprising given the impressive increase in depression research in recent years, which has led to important improvements in models of depression, including cognitive models, regarding the risk for the onset and maintenance of this disorder. This is not by any means unique to CBT; Joiner and Timmons (2009) note that proponents of interpersonal psychotherapy (IPT) have yet to incorporate several basic findings from interpersonal research. Indeed, the majority of treatment–outcome research on IPT has used existing IPT protocols (Markowitz & Weissman, 2012).

PHENOMENOLOGY

As described earlier, we first review the phenomenology of depression. Toward this goal, we describe the clinical symptoms, associated clinical features, prevalence, course and outcome, and common comorbid conditions. We then discuss the clinical implications of these findings.

Typical Symptoms

Depression is typified by low mood and/or loss of interest or pleasure in daily activities. People with MDD also experience cognitive, affective, and somatic difficulties, such as weight/appetite disturbance, sleep disturbance, psychomotor agitation/retardation, fatigue, feelings of worthlessness or guilt, and difficulty concentrating. Thoughts of suicide or death are also present in MDD at rates higher than those in the general population (as reviewed by Klonsky, Saffer, & Bryan, 2018).

The diagnostic criteria place much emphasis on vegetative or somatic symptoms of depression, but it is important to note that these symptoms are not specific to depression and are observed in many disorders, such as generalized anxiety disorder (GAD). There is evidence, however, to support the view that these symptoms are important to diagnosis but may not be central to the disorder. Mitchell, Goodwin, Johnson, and Hirschfeld (2008) found that somatic symptoms, when present, have high rule-in value for diagnosis but do not provide adequate classification on their own. Instead, according to these authors, mood and emotional symptoms are most clinically useful for making accurate diagnoses. This suggests that somatic symptoms play a role in depression but may not be central to the disorder in all cases. That being said, somatic symptoms do have prognostic significance; they are associated with more severe depression and predict greater depressive symptoms 6 months and 2 years later (Hung, Liu, Wang, Juang, & Yang, 2010).

However, what seems to particularly define depression across heterogeneous presentations is that it is a disorder of mood or affect. As a consequence, many newer etiological models of depression emphasize difficulties in emotion regulation as a central feature of the disorder (LeMoult & Gotlib, 2019; Nolen-Hoeksema, Wisco, & Lyubomirsky, 2008). Thus, we discuss emotion regulation in more detail in the etiology section of this chapter.

Associated Clinical Features

Depression is associated with several common clinical features not included in the diagnostic criteria. Anger or hostility is often a characteristic of depression (e.g., Painuly, Grover, Gupta, & Mattoo, 2011). Importantly, it seems that irritability is correlated with higher severity of depression (Perlis et al., 2005) and that it represents an additional risk factor within this population rather than a subtype of irritable depression. Others have also found that anger distinguished those with a history of a suicide attempt from those without one (Jandl, Steyer, & Kaschka, 2010; Perlis et al., 2005). Moreover, anger also appears to characterize depressed individuals' relationships with others, including friends, strangers, therapists, and spouses (Coyne, Burchill, & Stiles, 1991; Joiner, 2002). It is notable in this context that irritability is listed in the DSM as a symptom of depression in children and adolescents, but not in adults.

Interpersonal problems associated with depression are not restricted to hostility. Depressed individuals show social skills deficits and tend to seek both excessive reassurance and negative feedback from others. These aversive interpersonal behaviors can lead to frustration in close others and eventually, to rejection. Moreover, evidence from longitudinal studies indicates that excessive reassurance seeking and negative feedback seeking are robust risk factors for depression (Evraire & Dozois, 2011; Joiner & Timmons, 2009). Considering such interpersonal difficulties, it is not surprising that depression has been associated with elevated levels of marital dissatisfaction and discord, which appear to be both causes and consequences of depression (Davila, Stroud, & Starr, 2009; Joiner & Timmons, 2009).

The most serious associated feature of MDD is the increased risk for suicide. While death by suicide is rare, the consequences are dire. Depressive disorders are among the psychiatric disorders associated with the highest risk for suicide (as also reviewed by Bryan & Klonsky, Chapter 15, this volume). The likelihood of attempting suicide increases with depression severity, and acute and intense negative mood tends to precede attempts (Berman, 2009). Recent research has identified risk factors associated with suicidal ideation and attempts (Klonsky et al., 2018). Klonsky and May's (2015) three-step theory of suicide posits that the transition from suicide ideation to attempt occurs in three steps. In Step 1, the combination of pain (often psychological pain) and hopelessness leads to suicidal ideation. In Step 2, ideation escalates if pain exceeds level of connectedness with others, with valued roles, or with a sense of meaning and purpose. In Step 3, this can escalate to a suicide attempt if the individual has the capacity to attempt suicide. This includes dispositional capacity (e.g., high pain threshold, low fear of death), acquired capacity (habituation to pain, fear, or death through exposure to life events involving these themes), and practical capacity (knowledge and access to lethal means). Importantly, psychological pain, hopelessness, and low connectedness are all common features of depression.

Individuals with depression also display significant distress and/or impairment across domains of functioning. For example, depression is associated with sexual dysfunction, as well as relationship or marital problems (see Sbarra, Manvelian, Salinger, & Whisman, Chapter 12, and Pukall & Bergeron, Chapter 13, this volume). It is also frequently noted that occupational difficulties, including unemployment, absenteeism, and reduced work performance, are common in depression and are associated with greater symptom severity (e.g., Birnbaum et al., 2010).

Medical conditions are frequently comorbid with MDD. While there does not seem to be a consensus regarding whether (or which) medical conditions predict the occurrence of depressive episodes, numerous prospective studies have shown that depression increases the likelihood of physical illness (e.g., Salaycik et al., 2007). While depression may lead to physical illness as a result of engagement in maladaptive health behaviors (e.g., sedentary lifestyle, poor nutrition), there is also strong evidence that depression is associated with higher levels of systemic inflammation. Over time, high levels of inflammation may not only promote depressive symptoms but also result in the physical diseases commonly comorbid with depression (e.g., arthritis, cardiovascular disease, cancer; Slavich & Irwin, 2014). When a medical condition co-occurs with depression, the course of the medical condition is often longer, and the condition may complicate treatments for depression (Carney & Freedland, 2017; Freedland & Carney, 2009). Given its effects on physical health, it follows that depression also impacts all-cause mortality rates, particularly among older individuals. Evidence suggests that the mortality rate for people over age 55 increases three to four times with depression, and that mortality is substantially higher among individuals with depression during the first year living in a nursing home (American Psychiatric Association, 2013).

It should also be noted that cognitive factors are closely associated with depression. As described in the section on etiology, some cognitive content and processes have been identified as risk factors for depression (e.g., biased information processing), whereas others (e.g., negative automatic thoughts) appear as features or symptoms of current depression episodes.

Epidemiology

Depression is highly prevalent, with lifetime prevalence rates varying between 10 and 21% (Hasin, Sarvet, & Meyers, 2018). The 12-month and lifetime prevalence rate is between 13 and 26% for women, and between 7 and 15% for men (Hasin et al., 2018). Prevalence rates differ considerably by age group, with higher prevalence rates in individuals ages 18–64 years compared to individuals 65 years or older (Hasin et al., 2018). Interestingly, there has been a steady increase in prevalence rates of depression since World War II (Kessler et al., 2003). Although there are some discrepancies, most studies indicate that prevalence rates differ among people who identify as White, Asian, Native American, Hispanic, and Black (e.g., Hasin et al., 2018). For example, Hasin and colleagues (2018) reported higher prevalence rates in individuals who identify as White or Native American compared to individuals who identify as Asian, Hispanic, or Black in the United States. Higher rates of depression are also found among people living in poverty or who have less education (Kessler et al., 2003). Furthermore, the manifestation of depression symptoms can vary across cultures and ethnic groups. For instance, somatic complaints are more common in some non-Western populations, highlighting how culture can dramatically influence how clients present their complaints (American Psychiatric Association, 2013).

Another key finding is that women are 1.5 to 3 times as likely as men to experience depressive episodes (American Psychiatric Association, 2013). The gender difference in rates of depression begins to present around age 13 (Hankin et al., 1998), and the likelihood of developing depression is twofold for females compared to males by age 15 (Cyranowski, Frank, Young, & Shear, 2000). This gender gap may be

influenced by pubertal maturation differences between females and males, and the hormonal changes that occur during this time (e.g., changes in gonadal hormones; Cyranowski et al., 2000). However, the biological changes during puberty are not solely responsible for depression onset, as they interact with an increase in negative life events that occur during this developmental period (Cyranowski et al., 2000). A critical explanation for gender differences in the prevalence of depression is that differences in emotion regulation make females more prone than males to respond to major life events with depressive episodes (Jose & Brown, 2008; Nolen-Hoeksema et al., 2008). We return to this explanation later in the chapter when we discuss etiological models of the disorder.

Course

The first onset of a major depressive episode can occur at any age, with data from prospective studies suggesting that the first onset of depression typically occurs during adolescence (Kim-Cohen et al., 2003). An earlier age of onset is associated with greater impairment across several domains (e.g., social and occupational functioning, quality of life, suicide attempts, recurrence; e.g., Zisook et al., 2007). Importantly, individuals who develop depression frequently report a history of depressive symptoms even before their first episode. Put another way, subclinical forms of depression are very common and may predict the onset of clinical episodes of the disorder (Jeuring, Huisman, Comijs, Stek, & Beekman, 2016).

Moreover, depression can be both a chronic and recurrent disorder. The typical time to recovery from a depressive episode ranges from 3 to 6 months; however, a proportion of individuals continue to meet criteria for a depressive episode a year after initial diagnosis (Whiteford et al., 2013). In a recent study, approximately 37.6% of a depressed sample experienced a subsequent depressive episode within 4 years of remission (Scholten et al., 2016). Research shows that the probability of onset of another episode of depression increases by 16% with every episode a person experiences (Solomon et al., 2000).

Even though initial success rates of treatment are high, relapse rates also remain high; thus, there is considerable room to improve treatment effectiveness (Steinert, Hofmann, Kruse, & Leichsenring, 2014). For example, whereas Hollon, Stewart, and Strunk (2006) reported less than a 5% relapse rate 6 months after CBT for most anxiety disorders, they reported an almost 20% relapse rate 6 months after discontinuing cognitive therapy for depression. This is despite the fact that booster sessions were included in the depression, but not in the anxiety, treatment protocol. Furthermore, relapse rates may be worse for individuals who do not fully benefit from treatment. In a study of inpatients with depression, Conradi, Ormel, and de Jonge (2011) found that residual symptoms during remission were highly prevalent. Indeed, approximately one-third of patients reported residual symptoms during remission, and these symptoms enhanced risk for short-term relapse, poor social functioning, and poor health outcomes (Kennedy & Foy, 2005).

Comorbidity

Comorbidity among psychiatric disorders has received increasing attention in both clinical research and practice. The presence of comorbidity is now frequently

accepted as the rule rather than the exception. Moreover, the presence of comorbidity has important implications for the course, prognosis, and treatment of depression.

Results from the National Comorbidity Survey Replication indicate that 45% of individuals with psychopathology meet criteria for more than one disorder (Kessler et al., 2005). Of those with MDD, the percentage of people with comorbidity increases to 76%. There are particularly high correlations among MDD, GAD, social anxiety disorder (SAD), and agoraphobia. Within an outpatient sample with depression, for example, anxiety disorders (particularly SAD at 29.3% and GAD at 20.8%) were the most common comorbid disorders (Rush et al., 2005). Additionally, posttraumatic stress disorder (PTSD) and depression are commonly comorbid, with approximately 52% of individuals with PTSD presenting with comorbid MDD (Ikin, Creamer, Sim, & McKenzie, 2010; Rytwinski, Scur, Feeny, & Youngstrom, 2013). Individuals in the military and those who have experienced interpersonal trauma show higher rates of comorbid MDD and PTSD compared to the general population and survivors of natural disasters (Rytwinski et al., 2013). Concordance estimates are also high between lifetime MDD and substance-related disorders (Hasin, Goodwin, Stinson, & Grant, 2005). Finally, personality disorders, especially borderline personality disorder, are also highly associated with depression (Lenzenweger, Lane, Loranger, & Kessler, 2007).

The onset, frequency, and duration of depressive episodes are influenced by the presence of comorbid psychiatric disorders. For example, comorbidity of depression and anxiety disorders is associated with earlier onset and longer depressive episodes compared to the presence of depression alone (Penninx et al., 2011). Interestingly, symptoms of depression are also a risk factor for developing symptoms of anxiety, suggesting a bidirectional association between anxiety and depression (Jacobson & Newman, 2017). There is also a bidirectional association between PTSD and MDD; however, PTSD typically precedes depression (see Stander, Thomsen, & Highfill-McRoy, 2014, for a review). Similarly, the majority of longitudinal epidemiological data indicate strong evidence for a bidirectional association between depression and substance use. Heavier alcohol use and more frequent drug use during periods of childhood, adolescence, or early adulthood predicts the development of MDD (Brook, Brook, Zhang, Cohen, & Whiteman, 2002). Other studies have suggested that MDD might also be a precursor to substance use disorders (Armstrong & Costello, 2002; Rao & Chen, 2008). Regardless of temporal sequencing, the presence of substance use disorders is associated with more prolonged and recurrent depressive episodes.

When comorbidity is present, it is strongly related to the severity of symptom presentation. Comorbidity of depression and anxiety, in particular, has been associated with more severe complaints, increased behavioral avoidance, greater impairment in social and occupational functioning, higher risk of suicide, and higher utilization of mental health services (as reviewed in Belzer & Schneier, 2004). Of the anxiety disorders, the presence of depression with comorbid SAD or GAD is associated with particularly poor outcomes. Comorbidity of SAD and MDD is generally associated with increased risk for suicidality and alcohol dependence (Belzer & Schneier, 2004). Furthermore, data from an epidemiological study indicate that the presence of depression and comorbid GAD is highly disabling, over and above that of depression alone (Hunt, Slade, & Andrews, 2004). Those with comorbid PTSD and depression report reduced life satisfaction, diminished quality of life, and higher symptom severity compared to those presenting with either disorder alone (Ikin et

al., 2010). The co-occurrence of addictive disorders and depression is also associated with greater impairment, including more severe substance-related problems, increased frequency of behavioral problems, and more severe impairment in family, school, and legal domains (see Rao & Chen, 2008, for a review).

The co-occurrence of depression and other psychiatric disorders has been explained by a variety of factors, including cognitive biases, biological dysregulation, and coping strategies. Depressed individuals with a comorbid anxiety disorder show different attention and memory biases than do those without comorbidity (Dannlowski et al., 2006), as detailed in the section on etiology. Biased processing of negative information may place some individuals at risk for both depression and anxiety disorders. In addition, some anxiety disorders (e.g., SAD or agoraphobia) may lead to decreased social support, which increases risk for the onset of a depressive episode following stress. Researchers posit that the bidirectional association between PTSD and MDD is due to shared vulnerability factors (genetics, family environment) that increase the likelihood of the onset of both disorders (Stander et al., 2014). Depression is also associated with impaired executive functioning (Chakrabarty, Hadjipavlou, & Lam, 2016), which may preclude the ability to effectively manage trauma-related cognitions and may worsen prognosis of PTSD (Olff, Polak, Witteveen, & Denys, 2014). Other research has documented that increased risk for substance use disorders in depressed individuals might be partially accounted for by increased activation in the hypothalamus–pituitary–adrenal (HPA) axis (Rao, Hammen, & Poland, 2009). Substance use may also be an attempt to improve mood, particularly for individuals who do not use more adaptive emotion regulation strategies.

Clinical Implications

Research findings on the phenomenology of depression have various clinical implications that are relevant to therapists, irrespective of their theoretical orientation. To begin with, while psychopathology research indicates that low mood and diminished pleasure are the most characteristic features of this disorder, it also suggests that the presence of somatic symptoms, including changes in sleep, appetite, and psychomotor changes, is indicative of more severe and chronic depression. Because these symptoms do not always receive as much emphasis in the psychotherapy literature as psychological symptoms (e.g., feelings of worthlessness, guilt), vegetative or somatic features of depression may not always be on the therapist's "radar" despite their prognostic relevance. Therapists should assess these somatic symptoms when forming a case conceptualization and evaluating their clients' progress. The presence of somatic symptoms may inform the need for a greater number of sessions focused on interventions that target somatic symptoms and disengagement. For example, a cognitive-behavioral therapist may work through a more graduated hierarchy of activity scheduling as part of behavioral activation when working with a patient with vegetative symptoms to ensure that goals are realistic, achievable, and reinforcing. Across all forms of psychotherapy, therapists should assist patients who have difficulty with sleep or low energy with establishing a routine and may need to provide psychoeducation on sleep hygiene.

It is also important for therapists to remember that even though anger is not part of the diagnostic criteria, it is likely to be present and to manifest in the therapeutic relationship with a depressed client. Since the psychotherapeutic setting can be

viewed as a "microcosm" (Goldfried & Davison, 1976) of the client's ways of reacting with others, we can predict from basic research that some depressed clients will feel frustration toward treatment and therapists. Such frustration may lead to, and be exacerbated by, negative responses (at times referred to as "countertransference reactions") that have been observed across various theoretical orientations (Castonguay & Hill, 2017). When such interactions emerge, clinical judgment is needed to discern how much of this is a manifestation of the client's disorder and how much is a reflection of an alliance rupture (e.g., due to a lack of fit between the client's needs and the treatment implemented by the therapist, or the result of empathic failures on the part of the clinician).

Irrespective of the causes of the client's anger or hostile transactions, process-oriented discussions about maladaptive interpersonal behaviors may be instrumental to meaningful therapeutic change. Potential growth opportunities include increasing the client's awareness of the reactions the therapist and others have to the client's verbal and nonverbal communication style, as well as providing a model of how to recognize and discuss one's own negative emotions and their contributions to the dynamics of relationships (which scholars such as Safran & Segal, 1990, have referred to as "meta-communication"). As we discuss later in this chapter, other interpersonal issues related to the onset and maintenance of depression may merit being a foci of treatment.

When treating depressed clients, therapists must closely assess and monitor clients' risk for suicide. Several factors have been associated with suicide attempts and deaths, such as comorbid substance abuse and older age. In addition, specific symptoms and comorbid diagnoses are highly associated with suicide in depressed individuals, including anxiety, panic attacks, anhedonia, alcohol use, difficulty concentrating, and global insomnia (Klonsky, May, & Saffer, 2016). These should be carefully assessed before and during treatment with depressed clients. Furthermore, given findings from the three-step theory of suicide (Klonsky & May, 2015), pain (including psychological pain), hopelessness, and connectedness also need to be monitored and treated. Capacity to attempt suicide should be assessed, including dispositional (e.g., high pain tolerance, low fear of death), acquired (e.g., history of physical abuse, nonsuicidal self-injury, combat training) and practical (knowledge, expertise, and access to lethal means) contributors.

Therapists also need to assess nonsymptomatic aspects of client functioning, such as work performance and satisfaction, sexual functioning, health problems, and health behaviors. Treatments that target affect may lead to a decrease in depressive symptoms, but they may not directly or immediately impact these other dimensions, which can have critical effects on a person's quality of life. Unless these domains are addressed by specific therapeutic interventions or referral to other services, the scope of the therapist's interventions may be limited. While clients may experience better mood at the end of short-term therapy, they may have a recurrence of depression if they continue to be confronted with serious occupational and health difficulties—as suggested by basic research on stressors associated with the onset of depressive episodes.

In line with increasing the specificity of psychological treatments for depression, clinicians would be wise to familiarize themselves with cultural differences in the manifestation and report of depressive symptoms. This has clear implications for the possibility of over- or underdiagnosing individuals due to cultural insensitivity, and

could also have important impacts on treatment process, fit, and ultimately, effectiveness. For instance, establishing a shared understanding of the causes of somatic symptoms may be key in building a therapeutic alliance, and could provide a focus for monitoring treatment progress.

Basic psychopathology research also suggests that clinicians would be well advised to ask when clients experienced their first depressive episode, as well as to assess for other disorders frequently associated with depression (e.g., anxiety, substance abuse, and personality disorders) to inform treatment planning. As mentioned earlier, both early onset of depression and comorbidity have been linked with greater impairment. Because depressed clients with higher levels of impairment have also been shown to require longer-term therapy to achieve significant symptom reduction (as reviewed by Beutler, Blatt, Alimohamed, Levy, & Angtuago, 2006), an a priori limited number of sessions, which is frequently mandated in clinical trials and managed care settings, may not be sufficient for these clients. Comorbidities may also need to be addressed as part of treatment and, based on the case conceptualization, may need to be included in the treatment plan. Newer transdiagnostic therapies, such as the unified protocol for emotional disorders (Barlow et al., 2010) and group transdiagnostic CBT (tCBT; Norton, 2012), are designed to simultaneously treat multiple and comorbid disorders, and these should be considered when selecting interventions and planning treatment. Transdiagnostic empirically supported treatment (EST) provides an evidence-based framework for effectively targeting common mechanisms underlying multiple disorders, and represents a promising advance in psychotherapy research.

Careful assessment of the idiographic course of the disorder for a particular client is also informative for treatment planning. Basic research demonstrates that people experience a number of symptoms before suffering from a full episode of depression, and that having such an episode is a strong predictor of future recurrences. Subthreshold levels of depression should therefore be an important target of prevention and intervention efforts. Furthermore, considering the strong relationship between the number of previous episodes and the likelihood of future episodes, clinicians should assess their clients' past episodes of major depression. Since repeated episodes may indicate extreme vulnerability, therapists should expect that stronger (more intense, longer) doses of treatment might be necessary to foster restoration and/or consolidation of adaptive ways of functioning than the amount of therapy typically delivered in current EST for depression. Irrespective of the length and intensity of the treatment required, data also suggest that treatment should continue until symptoms fully abate; failure to do so may predict relapse.

Given that a substantial number of untreated individuals with depression do not experience remission from their symptoms for a year or longer, treatment should not be delayed with the hope that depression will go away by itself (i.e., spontaneous recovery). This recommendation for treatment can and should be made strongly by therapists, government agencies, and managed health care organizations because several psychological (and pharmacological; see Nathan & Gorman, 2021) treatments have received empirical support for their efficacy. In a review, Follette and Greenberg (2006) categorized three treatments as "well established" (behavior therapy, cognitive therapy, and IPT); two as "probably efficacious" (short-term psychodynamic therapy, process–experiential therapy); and one (mindfulness therapy) as "experimental," based on promising initial results. The evidence supporting the effectiveness of

these different treatment orientations for depression means that it may be clinically unsound for clinicians not to use ESTs, at least as first lines of intervention.

Follette and Greenberg (2006) also derived six principles of change from treatment manuals of empirically supported psychological therapies that could guide evidence-based practice for depression: (1) challenging cognition and behavior; (2) increasing positive reinforcements and decreasing negative reinforcements (for avoidant and depressotypic behavior) in the client's life; (3) improving the client's interpersonal functioning; (4) improving marital, family, and social environment; (5) fostering emotional awareness, acceptance, and regulation; and (6) providing a treatment that is both structured and focused. These therapeutic strategies cut across several ESTs and may be incorporated into most forms of psychological treatments. Moreover, these principles are precise enough to guide the focus of clinicians' interventions without being restricted to a narrow and orientation-specific set of prescribed procedures (see Castonguay & Beutler, 2006). There are, for example, many techniques (e.g., interpretation, cognitive restructuring) that therapists can use to challenge client appraisals or views of self and others, all with similar aims (Goldfried, 1980). In addition, principles of change can help therapists to increase their repertoire of interventions and individually optimize treatment if their treatment protocol does not sufficiently target factors believed to cause or maintain depression for that particular client. For example, in addition to using effective techniques to address "internal" or intrapersonal issues (e.g., exploring a client's emotions or challenging distorted cognitions), experiential, psychodynamic, and cognitive-oriented therapists should also assess and, if necessary, address interactional issues that are frequently associated with depression, such as social isolation, interpersonal skills deficits, and dysfunctional marital or familial relationships. Detailed descriptions of how these and other empirically derived principles of change are implemented in routine treatment of depression (both with or without comorbid substance abuse or personality disorder) by experienced therapists of different theoretical orientations have been recently presented (Castonguay, Constantino, & Beutler, 2019).

As mentioned earlier, however, not all depressed clients benefit from therapy, and many who do benefit fail to retain their therapeutic gains after the completion of treatment. There is much room for improvement in our interventions for this debilitating and costly disorder. While it is unlikely that devising new forms of therapy is the best strategy to improve treatment outcomes, a more fruitful strategy may be to build on and improve already established and supported treatments. This could be done by modifying existing therapies based on the findings of psychotherapy process research. As an example, promising evidence suggests that integrating interpersonal and humanistic interventions aimed at repairing alliance ruptures (which are robustly linked to worse outcomes in depression treatment) may improve the efficacy of cognitive therapy for depression (Castonguay et al., 2004; Constantino et al., 2008; Safran & Muran, 2000). Other researchers have found that therapists are more effective when they promote their client's autonomy by encouraging them to set therapeutic goals and make decisions (Zuroff, Koestner, Moskowitz, McBride, & Bagby, 2012).

As most clinicians are already aware, depression is one of the most common problems encountered in clinical practice. Interestingly, and perhaps alarmingly, research has shown that the prevalence of depression may be increasing (Hidaka, 2012). According to Seligman (1989), this increase points to social/environmental

determinants of depression. Contrasting a 10-fold increase in prevalence observed in the general population with the rate of depression in some "nonmodern" cultures (e.g., Amish), he argued that contemporary society's emphasis on individualism rather than on the common good plays a causal role in depression. It may be that the lack of commitment to common goals has reduced opportunities for group activities and a sense of belongingness that buffer against depression when one is confronted with personal difficulties or failures. An overinvolvement in activities aimed at increasing individualistic accomplishments, wealth, and comfort might make it more difficult to ask others for help when we are experiencing major difficulties. With this in mind, and irrespective of their orientation, clinicians are encouraged to help clients renew or create meaningful relationships. They should teach clients skills to be appropriately vulnerable, and they should encourage clients to become involved in purposeful and pleasurable activities with others. These clinical recommendations focused on improving clients' interpersonal functioning, as well as improving the marital, family, and social environment, emerge from the basic research described earlier and are consistent with the principles of change derived from ESTs. Such a convergence of findings across different domains of research may not be a surprise considering the core aspects of human functioning to which they appear to be linked. As argued by MacLean (1985), communication and attachment are two of the evolutionary developments that differentiate humans, and all mammals, from reptiles. Denying or not attending to such ways of being forces ourselves and our clients to fight a losing evolutionary battle!

Helping clients to be more engaged interpersonally, however, may be more difficult with some people than with others. Individual preferences and cultural norms vary in terms of agency and communion needs and tendencies. As described by Bonanno and Castonguay (1994), when conducting therapy with agency-oriented people, it might be best to initially focus on intrapersonal or individualistic issues, so that the tasks and goals of therapy may be more synchronized with individuals' typical ways of coping with life difficulties. With the development of a stronger therapeutic alliance, however, we may begin to focus more on what are likely to be deficits in the interpersonal functioning of these individuals.

ETIOLOGY

As we mentioned earlier in this chapter, the extent of the empirical literature on the determinants of depression prevents us from providing a comprehensive review of vulnerability factors for this disorder. Among the most obvious omissions in the following subsections are basic findings related to some biological models (e.g., neurochemical abnormalities, genetic disposition) and learning (e.g., operant and vicarious conditioning) that have been associated with depression (for reviews, see Burke, Davis, Otte, & Mohr, 2005; Johnson, Joormann, LeMoult, & Miller, 2009; Sullivan, Neale, & Kendler, 2000). These well-established vulnerability factors have provided a foundation for some of the currently supported psychological and pharmacological treatments. Our goal, however, is to focus primarily on recent basic findings and etiology research that have not yet been sufficiently incorporated in our current treatments.

Cognitive Theories

Cognitive models of depression posit that depressed individuals exhibit cognitive biases in all aspects of information processing, including memory, attention, and interpretation (Mathews & MacLeod, 2005). These cognitive biases may increase risk for the onset and maintenance of depressive episodes. Although empirical results for some of these biases are mixed, others have been consistently associated with depression and may have important treatment implications.

Memory

Overall, there is strong evidence for biased memory in depression (Mathews & MacLeod, 2005). In fact, better recall of negative relative to positive information represents one of the most robust cognitive findings associated with major depression. Memory biases are found most consistently in free recall and explicit memory tasks. For example, when attempting to recall details of past events or recently viewed words, individuals with depression are more likely than nondepressed controls to recall negative or unpleasant information.

Additional evidence that individual differences in memory play an important role in depression comes from research on overgeneral memory (Williams et al., 2007). These studies show that depressed participants respond to positive and negative cues with memories that summarize a category of similar events. For example, when asked to describe a specific positive memory, nondepressed controls provide specific details of a past event (e.g., "the day my family went to Lake Michigan"); however, individuals with depression tend to recall a more general period in time (e.g., fifth grade). Importantly, this research has demonstrated that overgeneral memories are associated with poor problem solving, difficulty imagining specific future events, and delayed recovery from episodes of depression. Moreover, overgeneral memories remain stable outside of episodes of the disorder and have been shown to predict later onset of depressive episodes following life events.

Attention

Researchers have also investigated whether depressed individuals display biases in attention, which refers to the tendency to visually attend to negative versus positive information in the environment (see LeMoult & Gotlib, 2019, for a review). Although evidence for attentional biases in depression has been mixed, there are specific circumstances when depressed individuals do display attentional biases, and understanding these circumstances can offer treatment-relevant insights. For example, whereas depressed and nondepressed individuals do not differ in their attention to negative information presented for short durations, depressed individuals spend *more* time attending to negative information that has captured their attention. These findings suggest that depressed individuals do not automatically orient their attention toward negative information in the environment, but once such information becomes the focus of their attention, they have greater difficulty disengaging from it. Notably, difficulty disengaging attention from negative stimuli has important implications, such as predicting greater reactivity to stress (Sanchez, Vazquez, Marker,

LeMoult, & Joormann, 2013) and a worse course of depression symptoms over time (Disner, Shumake, & Beevers, 2017).

Interpretation

Results also have been mixed regarding whether depression is characterized by an interpretation bias (Lawson, MacLeod, & Hammond, 2002). Recent evidence, however, suggests that those with depression typically make negative interpretations of ambiguous information, and that symptom severity correlates with the magnitude of their negative interpretation bias (see LeMoult & Gotlib, 2019, for a review). In a meta-analysis, Everaert, Podina, and Koster (2017) found that negative interpretation biases are stronger for self-referential stimuli (i.e., referencing personal experience) than for non-self-referential stimuli. Other research indicates that healthy control participants show a positive cognitive bias that does not appear in participants with depression. Thus, whereas healthy controls have a positive interpretation bias, individuals with depression may have a more "realistic" view of the world (McKendree-Smith & Scogin, 2000).

Cognitive Control

Recent cognitive theories posit that the aforementioned depressogenic biases in attention and memory are due in large part to biases in cognitive control (Joormann, 2010). Indeed, depression is associated with difficulty in multiple aspects of cognitive control, including difficulty inhibiting negative information from entering working memory, difficulty shifting away from negative information held in working memory, and difficulty disengaging from negative material in working memory (LeMoult & Gotlib, 2019). Consequently, depressed individuals exhibit sustained processing and increased elaboration of negative content, often observed as biases in attention and memory (described earlier) or as rumination (described in more detail later in this chapter). Because the experience of negative mood states and negative life events is associated with the activation of mood-congruent cognitions in working memory, the ability to control the contents of working memory may be critical in differentiating people who recover easily after experiencing negative affect from those who initiate a vicious cycle of increasingly negative ruminative thinking and deepening sad mood.

Summary

Overall, the literature points to several cognitive biases that may have important treatment implications in depression. For example, as described earlier, depressed individuals have difficulty disengaging from negative stimuli once it enters into their awareness (e.g., Joormann & Gotlib, 2007), and this may lead to the increased elaboration of negative information and the negative memory biases frequently observed in depression (Mathews & MacLeod, 2005). In addition, overgeneral memory in response to negative cues is thought to contribute to negative mood states and is considered a risk factor for the development of depression (e.g., Rawal & Rice, 2012). Moreover, evidence suggests that deficits in cognitive control might underlie

the attention and memory biases observed in depression. Such deficits might also contribute to increased elaboration of negative information and rumination (Joormann, 2010), which are key contributors to prolonged negative affect. As such, training individuals to control the contents of working memory could interrupt the cognitive mechanisms that contribute to depressive episodes.

Emotion Regulation in Depression

Depression has been referred to as a disorder of emotion regulation (ER; Gross & Muñoz, 1995; Joormann & D'Avanzato, 2010). Sustained negative affect and persistent reduction in positive affect are the hallmark features of a diagnosis of a major depressive episode. How can we explain why some people respond to minor stressors with increasingly negative affect that can devolve into a full-blown depressive episode, while others seem never to become depressed, even when they face major adversity? Individual differences in ER, the important ability to regulate affective states, may provide an answer to these questions (Bonanno, 2005). Specifically, depression-vulnerable and nonvulnerable people do not differ in their initial response to a negative life event. Both groups experience sadness and respond with the activation of mood-congruent cognitions and increased accessibility of mood-congruent memories. Depression-vulnerable and nonvulnerable people do differ, however, in their ability to recover from this initial response (Teasdale, 1988). This suggests that we need to examine factors that impair or support this recovery.

The construct of ER, which evolved from the more general concept of coping, involves the use of behavioral and cognitive strategies to influence the duration, intensity, and experience of affect (Thompson, 1994). Emotions are regulated for a variety of reasons, such as to enhance desirable feelings, maintain social norms, and foster interpersonal relationships (Fischer, Manstead, Evers, Timmers, & Valk, 2004). Theories of depression emphasize that increased risk for MDD stems from the use of strategies that fail to down-regulate negative emotions after their initial onset (e.g., LeMoult & Gotlib, 2019; Nolen-Hoeksema et al., 2008), which leads to prolonged negative affect (John & Gross, 2004). In response to distress, the use of less effective ER strategies (e.g., rumination, suppression) has been associated with multiple psychological disorders, including depression (Kring & Werner, 2004), whereas effective ER strategies (e.g., reappraisal, emotional disclosure) are typically characteristic of healthy psychological functioning and mood (Gross, 1998; Nils & Rimé, 2012). In fact, studies indicate that less frequent use of effective ER strategies such as reappraisal is not only characteristic of those with depression, but it also predicts an increase in depressive symptoms (Dryman & Heimberg, 2018). These findings support the claim that more frequent use of certain maladaptive strategies (e.g., emotion suppression, rumination, and catastrophizing) and less frequent use of other, more adaptive strategies (e.g., reappraisal, self-disclosure) are related to depression (e.g., Garnefski & Kraaij, 2006; Kahn & Garrison, 2009).

An emerging line of research has focused on ER *flexibility,* the ability to flexibly use ER strategies to meet situational demands (Aldao, Sheppes, & Gross, 2015). This research deemphasizes the uniform effectiveness of ER strategies (described earlier) and instead argues that ER strategies are adaptive when they fit the context and facilitate goal pursuit (Aldao et al., 2015; Bonanno & Burton, 2013). For instance,

although reputed to be uniformly effective, reappraisal is less effective at regulating emotions elicited by stressors that are controllable by the individual. In this instance, engaging with and solving the problem itself is more effective (Bonanno & Burton, 2013). Indeed, research indicates that higher ER flexibility is associated with reduced symptoms of psychopathology (Aldao et al., 2015).

In addition, studies suggest that impaired ER not only characterizes currently depressed people but also may be observed after recovery. For example, Ehring, Fischer, Schnülle, Bösterling, and Tuschen-Caffier (2008) found that remitted depressed participants had greater difficulty regulating their negative emotions and more frequently used rumination and catastrophizing compared to control participants. Within the depression literature, rumination and distraction have received the most attention; however, emerging evidence also points to the important role of suppression and reappraisal.

Rumination

Rumination is a process of responding to distress by repetitively and passively focusing on symptoms of low mood and the potential causes or consequences of these symptoms. It is differentiated from negative automatic thoughts because it is a *style* of thinking rather than being characterized solely by negative thought *content* (Nolen-Hoeksema et al., 2008). This ER strategy is a particularly salient risk factor for depression because it exacerbates and prolongs depressed mood (see review by Nolen-Hoeksema et al., 2008). Empirical evidence has shown that rumination predicts the onset of major depressive episodes and higher levels of depressive symptoms, and that it mediates the gender difference in depressive symptoms (for a review, see Nolen-Hoeksema et al., 2008). Many studies that experimentally tested the effects of rumination show that induced rumination leads to sustained negative mood, increased negative cognitions, increased overgeneral autobiographical memory, and decreased problem solving in depressed participants (e.g., Nolen-Hoeksema et al., 2008).

Distraction

In contrast to rumination, "distraction" involves engaging in positive or neutral activities in order to divert one's thoughts from symptoms of distress and depression (Lyubomirsky, Caldwell, & Nolen-Hoeksema, 1998). For example, one may lower the level of subjective emotion experienced by focusing on thoughts unrelated to the situation (e.g., planning a supermarket shopping list). It is important to differentiate attentional distraction, which involves shifting visual or auditory attention away from emotion-producing stimuli, as discussed earlier, from cognitive distraction described here, which involves shifting one's thoughts away from the emotion or emotion-producing stimuli. Researchers have reported that dysphoric, clinically depressed, and recovered depressed people can use cognitive distraction to repair an induced negative mood state (Joormann & Siemer, 2004; Joormann, Siemer, & Gotlib, 2007). Indeed, distraction has been associated with adaptive outcomes such as faster physiological recovery from stress (Vickers & Vogeltanz-Holm, 2003), decreased depressed mood (Kuehner, Huffziger, & Liebsch, 2009), and shorter duration of

depressive symptoms (Nolen-Hoeksema, Morrow, & Fredrickson, 1993). However, depressed participants are less likely to use distraction even though they state that it could help them alleviate their negative affect (Lyubomisky & Nolen-Hoeksema, 1993). Importantly, while distraction may be adaptive for limited periods, persistent distraction is an ineffective long-term ER strategy because it hinders effective problem solving (Campbell-Sills & Barlow, 2007; Kross, Ayduk, & Mischel, 2005).

Suppression

Suppression comprises both thought suppression and expressive suppression. Whereas thought suppression involves attempting not to think of an emotion-provoking event or stimuli, expressive suppression involves inhibiting the outward expression of an emotional experience, including facial, postural, and verbal cues. Ironically, research has shown that thought suppression can lead to increases in the accessibility and frequency of the thought the individual is attempting to suppress (Kircanski, Craske, & Bjork, 2008), which in turn increases their emotional response (e.g., Campbell-Sills, Barlow, Brown, & Hofmann, 2006). Several studies suggest that depressed and recovered depressed people are more likely than control participants to use thought suppression, and that the use of this ER strategy may play an important role in the onset and maintenance of depression (Wenzlaff, Rude, Taylor, Stultz, & Sweatt, 2001). Kahn and Garrison (2009) documented that depression is associated with not only more suppression but also less emotional self-disclosure (Kahn & Garrison, 2009), a technique that has far-reaching benefits. Indeed, self-disclosure can free up cognitive resources, can buffer against the effects of stress, has long-term health benefits, and can serve as an emotionally therapeutic experience (Pennebaker, 1997a, 1997b).

Although studies reveal that participants can successfully engage in expressive suppression when instructed, participants also show greater physiological activation when doing so. This suggests that expressive suppression is an effortful means of regulating emotion (Gross, 1998). Expressive suppression has also been found to be ineffective at regulating negative feelings, and several studies indicate that it decreases the experience of positive, but not negative, emotions (e.g., Gross & John, 2003; Nezlek & Kuppens, 2008). Moreover, the habitual use of this strategy is associated with lower well-being and greater symptoms of depression (Haga, Kraft, & Corby, 2009; LeMoult & Gotlib, 2019). Even outside the laboratory, greater use of expressive suppression has been linked to lower social support, lower social satisfaction, and decreased closeness to others (Srivastava, Tamir, McGonigal, John, & Gross, 2009).

Reappraisal

Reappraisal has increasingly become a focus within the depression literature. This construct refers to the process of changing the way one thinks about a situation to alter its emotional impact (John & Gross, 2004). Some people may reappraise a situation by changing their point of view (i.e., taking a third-person rather than a first-person perspective), whereas others change their interpretation of the situation (i.e., viewing a difficult task as a challenge rather than as a threat; Ray et al., 2005).

Habitual use of this strategy has been associated with adaptive changes in emotions, cognitions, and physiological arousal. For example, reappraisal has been linked to increased well-being, fewer depressive symptoms, increased inhibition of negative material, greater positive autobiographical recall, more adaptive physiological reactivity to distress, and better interpersonal functioning (Gross & John, 2003; LeMoult & Gotlib, 2019). Compared to other ER strategies, such as expressive suppression, reappraisal is associated with the experience of more positive affect and less negative affect (Gross & John, 2003).

Neuroendocrinology

Stressful life events and other psychosocial stressors are robustly associated with the onset, severity, and course of MDD (see review in Hammen, 2005). Dysfunctions in the neuroendocrine system might partially explain the link between stress and depression, making "neuroendocrinology"—the study of interactions between the endocrine and the nervous system—increasingly important (LeMoult, 2020). A central component of the neuroendocrine system is the HPA axis, which is essential in regulating the hormone cortisol. Cortisol affects many aspects of the body, including metabolism, immune function, and the brain. In addition to being released in response to stressors, it is released spontaneously throughout the day and shows patterns of diurnal variation. Diurnal cortisol production is influenced both by sleep–wake cycles and by affective state (Dedovic & Ngiam, 2015).

Given that functioning of the HPA system is so integrally related to the human stress response, it is not surprising that atypical patterns of both diurnal HPA functioning and acute HPA stress reactivity have been documented in various psychiatric disorders, including depression (e.g., Porter & Gallagher, 2006). Investigating neuroendocrine response in these disorders is important because it might represent a crucial link to the neurobiological research we describe later. For example, the hippocampus and the prefrontal cortex (PFC) are the brain regions with the highest density of glucocorticoid receptors. Chronic exposure to stressful events that result in persistent and increased cortisol secretion might lead to neurotoxicity in these areas (Sapolsky, 2000). Chronic cortisol secretion may also disrupt functioning in regions of the brain that are responsible for regulating emotion (e.g., the PFC and the amygdala), which, in turn, can interfere with the ability to cope effectively with subsequent stressors.

Differences in basal HPA activity and diurnal cortisol fluctuations have been found in individuals with MDD compared to their healthy control counterparts. Specifically, compared to healthy control participants, individuals with depression tend to show an exaggerated cortisol awakening response (CAR) and elevated cortisol levels throughout the day (see reviews by Burke et al., 2005; Dedovic & Ngiam, 2015; Pariante & Lightman, 2008). In a meta-analysis, Burke and colleagues (2005) reported that depressed individuals exhibit less cortisol reactivity in response to stress and slower cortisol recovery poststress compared to nondepressed individuals. They also reported that the higher the cortisol levels at baseline, the less cortisol produced after the stressor. Whereas some researchers have reported that HPA axis abnormalities in depressed individuals are state-dependent and ameliorate with clinical remission (e.g., Ribeiro, Tandon, Grunhaus, & Greden, 1993), more recent research has

documented dysregulated HPA-axis function in those with a history of depression (LeMoult, Chen, Foland-Ross, Burley, & Gotlib, 2015; Lok et al., 2012). Those remitted individuals who continue to show abnormal cortisol profiles during periods of remission also have a higher rate of relapse (e.g., Aubry et al., 2007; Zobel et al., 2001).

Importantly, elevated evening cortisol levels have been found in depressed adolescents compared to nondepressed adolescents, and elevations in evening cortisol levels within the sample of depressed adolescents predicted a recurrent course of the disorder at a 7-year follow-up assessment (Mathew et al., 2003; Rao et al., 1996). In addition, numerous studies have indicated elevated cortisol secretion in response to acute stressors in depressed children (e.g., Luby et al., 2003; Watamura, Donzella, Alwin, & Gunnar, 2003). Elevated levels of cortisol have also been found in healthy youth at risk for depression based on a maternal history of the disorder (LeMoult, Chen, et al., 2015), and in turn, elevated cortisol has been shown to predict the first onset of depression in these youth (LeMoult, Ordaz, Kircanski, Singh, & Gotlib, 2015).

Neuroimaging

Brain areas and neural circuits associated with the generation and regulation of emotional states have received increasing attention in research on affective disorders (see Davidson, Pizzagalli, & Nitschke, 2002, for a review). Theorists have posited that emotional behavior is linked to the functioning of two neural systems, a ventral system and a dorsal system. The ventral system involves brain regions that are important for identifying the emotional significance of a stimulus and producing affect; the dorsal system is important for executive function and cognitive control, including selective attention, planning, and effortful regulation of affective states. Adaptive emotional behavior is posited to depend on the integrity and balanced interaction of these systems (Ochsner & Gross, 2005).

Although the ventral system includes many different brain structures, the vast majority of mood disorders research on this system focuses on the amygdala and the subgenual anterior cingulate cortex (ACC). The dorsal region that has received the most attention is the PFC, with a particular focus on the dorsolateral PFC (DLPFC), and some focus on the ventromedial PFC (VMPFC; e.g., Gotlib et al., 2010). Although we focus on these brain regions, it is worth noting that they interact with many different divisions of the ACC, the PFC, and other brain regions in important ways that have repercussions for understanding emotion. The amygdala has been shown to play a prominent role in emotionally mediated attention, in assigning emotional significance to stimuli, and in memory of emotionally significant events. The subgenual ACC appears to mediate the subjective experience of emotion and emotional reaction to stimuli, particularly in guiding reward-seeking behavior (Pizzagalli et al., 2004). The DLPFC appears to be involved in the regulation of emotion (Drevets, 2000). For example, it appears to diminish amygdala activity in response to an emotional challenge (Siegle, Steinhauer, Thase, Stenger, & Carter, 2002).

Depression is associated with dysregulated neural functioning, particularly with a hyperactive ventral system (more specifically, the amygdala and subgenual ACC) and a hypoactive dorsal system (DLPFC; see Mayberg, 2002), as well as deficits in areas associated primarily with the processing of rewarding stimuli (Davidson et al., 2002). For example, compared to controls, depressed individuals exhibit greater activity in the amygdala (e.g., Siegle, Thompson, Carter, Steinhauer, & Thase,

2007) and subgenual ACC in response to emotional stimuli (e.g., Gotlib et al., 2005). Depression is also associated with diminished top-down activation in areas of the brain responsible for regulating emotion (i.e., DLPFC; see Gotlib et al., 2010, for a review), which exacerbates excess bottom-up activation in the ventral system.

A concern, though, is that much of this research examines participants during episodes of MDD, when group differences in activation could reflect a state-dependent feature of depression rather than an etiological risk factor. To disentangle epiphenomena from causal mechanisms, studies examining at-risk and unaffected family members are particularly important. Although more extensive research is needed, preliminary studies indicate that compared to healthy controls, youth at risk for depression display increased activation of the ventral system (Gotlib & Hamilton, 2008) and decreased activation of the dorsal region (e.g., Joormann, Cooney, Henry, & Gotlib, 2012). These findings suggest that amygdala response to emotional stimuli in depression is not just a state marker, but rather a preexisting vulnerability factor that could precipitate disorder onset.

Given that anhedonia is a cardinal symptom of depression and that aberrant reward responsivity is therefore important for understanding MDD (Kasch, Rottenberg, Arnow, & Gotlib, 2002), researchers have also begun to attend to areas implicated in the anticipation and response to reward. These include the ventral striatum and the nucleus accumbens (e.g., Knutson, Fong, Adams, Varner, & Hommer, 2001), which are regulated by the PFC (e.g., Phillips, Drevets, Rauch, & Lane, 2003). Along these lines, researchers have found that depressed individuals exhibit reduced activation in the nucleus accumbens relative to controls (Pizzagalli et al., 2004). Moreover, deep brain stimulation of the accumbens and the ventral capsule/ventral striatum has been shown to significantly reduce symptoms of anhedonia in those with depression (Malone et al., 2009).

Interpersonal and Developmental Contributions

Interpersonal factors have been associated with the onset, maintenance, and recurrence of depression across different phases of development. Early risk factors for depression include several manifestations of inadequate parenting, childhood maltreatment, and interpersonal life stress (Humphreys et al., 2020; LeMoult et al., 2020). Caregiver deficits that are risk factors for later depression in children (both in youth and adulthood) include various types of neglect, such as inattentiveness, withdrawal, understimulation, emotional unresponsiveness, lax enforcement, inconsistent reinforcement, and low frequency of positive reinforcement. Experiences of severe neglect and deprivation are particularly common when children are reared in institutions such as orphanages (Gunnar, Van Dulmen, & International Adoption Project Team, 2007). Other forms of early life interpersonal stress and childhood maltreatment may involve aggression toward the child, such as overt hostility, intrusiveness, and antipathy, as well as emotional overinvolvement, overstimulation, exposure to maladaptive interpersonal behavior and affect, and physical and sexual abuse. Insecure parent–child attachment and parental conflict have also been identified as early vulnerability factors for developing depression. Similarly, losing a parent, particularly to suicide or an accident, and having a depressed parent are robust predictors of depressive disorders (Davila et al., 2009; Goodman & Brand, 2009; Hammen, 2009).

A number of these early risk factors frequently co-occur and act in tandem to cause later depression. Impaired parenting, for example, has been found to mediate the relationship between parental depression and child psychological difficulties (Goodman & Brand, 2009; Hammen, 2009). Furthermore, children who are exposed to greater cumulative adverse events experience more negative outcomes in adolescence (Appleyard, Egeland, van Dulmen, & Sroufe, 2005). It should also be noted that children's vulnerabilities or problems (biological, cognitive, affective, and behavioral) may contribute to parenting difficulties, for instance, by prompting criticism from parents (see Goodman & Brand, 2009; Hammen, 2009). Regardless of whether their impact is direct or indirect, early interpersonal adversities and maltreatment likely contribute to the development of difficulties (e.g., deficits in emotion regulation and interpersonal skills, social-cognitive biases, insecure attachments, dysfunctional attributions, and poor ability to respond to stress) that make children vulnerable to later depression (Goodman & Brand, 2009; Jopling, Tracy, & LeMoult, 2020; Rnic et al., 2018).

In addition to childhood adverse experiences, major stressful events across the lifespan have also been associated with depression. As reviewed by Monroe, Slavich, and Georgiades (2009), most of the major events that have been linked with the onset of depression are interpersonal, such as humiliation, social defeat, rejection, social exclusion, ostracism, and loss. Among the evidence reported by Monroe and colleagues on the role of social-environmental factors is the finding that depressed individuals who experience major stress before the onset of depression have more severe symptoms than depressed people who were not confronted with such stressors (Monroe, Harkness, Simons, & Thase, 2001; Tennant, 2002). Research also suggests that stressful events occurring during a depressive episode may interfere with recovery (Mazure, 1998). Although interpersonal stressors are most closely associated with an onset of depression, other noninterpersonal events, such as occupational failure or job loss, also predict the disorder. Though findings are somewhat inconsistent, evidence indicates that some individuals may be particularly vulnerable to noninterpersonal stressors due to personality or cognitive styles related to placing a strong emphasis on achievement and autonomy (e.g., Mazure, Bruce, Maciejewski, & Jacobs, 2000).

Stressful events occurring in adulthood may also be an important mechanism for the intergenerational transmission of depression. For instance, studies suggest that parental stressful life events may mediate the relationship between parental depression and psychological problems experienced by their children. For example, the chronically stressful conditions in which depressed individuals live can increase the prevalence of depression in offspring (Hammen, 2009).

Romantic difficulties are frequently experienced interpersonal risk factors for depression in adulthood. When romantic relationships are supportive, they can be protective against depression. However, depression is also associated with challenges in romantic relationships, including a lack of support from one's spouse, couples' difficulties with solving problems, lower relationship satisfaction, and dissolution in romantic or marital relationships (Davila et al., 2009). Importantly, the association between depression and romantic problems is bidirectional. Negative events or chronic stressors in close relationships (e.g., conflict, tension, or loss) place individuals at risk for depression. In turn, depressed individuals are more likely to

contribute to negative dynamics and events in their relationships (Davila et al., 2009), often via maladaptive interpersonal behaviors.

Other research supports the role of specific interpersonal behaviors/styles as risk factors for the onset and maintenance of depression. These include excessive reassurance seeking, interpersonal dependency, and an insecure attachment style in adulthood (Joiner & Meltalsky, 2001; Joiner & Timmons, 2009; Roberts, Gotlib, & Kassel, 1996), all of which result in negative interpersonal life events, including conflict and rejection by others. Depression is also associated with *negative* feedback seeking, a concept linked to the social-psychological concept of self-verification (Swann, 1983). Numerous studies have demonstrated that depressed individuals prefer to be with (and/or give more credibility to) people who react negatively to them compared to people who provide them with positive feedback (e.g., Giesler, Josephs, & Swann, 1996; Swann, Wenzlaff, Krull, & Pelham, 1992). The inferred mechanism is that the desire for safety in having one's self-concept confirmed (even when it is negative) is stronger than the disconfirming (but positive) impact that enhancement feedback can have. However, as noted earlier, depressed individuals also tend to excessively seek reassurance from others. Evraire and Dozois (2011) posit that this discrepancy in the types of feedback depressed individuals attempt to elicit is explained by a desire for *global* enhancement, attained through reassurance seeking, and verification of *specific* attributes, pursued via negative feedback seeking.

As can be evidenced in several of the previous examples, depression-prone individuals have characteristics and engage in behaviors that increase their exposure to stressors—particularly interpersonal stressors—that in turn further increase depression risk. This process is referred to as "stress generation" (Hammen, 2006), and it is a key mechanism through which the cycle of stress and depression intensifies over time.

Clinical Implications

Several clinical recommendations can be derived from the vulnerability factors we have described. As previously mentioned, cognitive and affective factors involved in the onset and maintenance of depression are closely interconnected. Interestingly, research investigating both cognitive biases and ER strategies suggests that depressed individuals become stuck in maladaptive patterns of automatic reactions. Studies on attention, interpretation, and memory indicate that depressed individuals have difficulty disengaging from negative information. Similarly, ER studies show that depressed individuals inflexibly use strategies that prevent them from recovering from the initial response of sadness and its related mood-congruent thoughts and memories. If these factors do play a causal role in the onset and maintenance of depression, therapeutic interventions that can directly modify them may be particularly promising.

Cognitive Biases

Interestingly, experimental studies suggest that it might be possible to modify depressed mood by modifying the cognitive biases posited to underlie it. Initial studies demonstrated that training depressed people to disengage their attention

from negative material improves mood and reduces reactivity to stressful events (as reviewed by Hertel & Mathews, 2011). Positive attentional biases can also be trained (Wadlinger & Isaacowitz, 2008). Moreover, positive attentional bias training reduces attention to negative images during a stress induction (Wadlinger & Isaacowitz, 2008), and reduces self-reported and cortisol responses to stress (Dandeneau, Baldwin, Baccus, Sakellaropoulo, & Pruessner, 2007). These findings provide an exciting example of the way modifying cognitive biases may remediate dysfunctions in the neuroendocrine system, and they are consistent with theoretical models of depression positing a bidirectional association between cognitive and biological dysregulation (LeMoult, 2020).

Encouraging results have also been documented for interpretation bias training (see Hallion & Ruscio, 2011). Researchers have found that adults and youth can be trained to positively interpret ambiguous information (Holmes, Lang, & Shah, 2009; LeMoult et al., 2018), and that doing so attenuates depressed mood and responses to stress (Hallion & Ruscio, 2011). For example, Holmes and colleagues (2009) demonstrated that training positive biases helped to alleviate a subsequently induced negative mood state. These results suggest that changes in interpretation biases can lead to changes in emotional responding.

Other training studies have targeted the negative memory biases often found in depression (LeMoult & Gotlib, 2019). Overall, findings suggest that it is possible to alter the negative and overgeneral memory biases associated with depression. In positive memory enhancement training (PMET), for example, depressed participants were asked to recall a happy memory in a way that facilitated vivid recall of the moment-to-moment details (Arditte Hall, De Raedt, Timpano, & Joormann, 2018). Arditte Hall and colleagues (2018) found that PMET enhanced the specificity of depressed participants' positive memories and facilitated mood repair following a sad mood induction. Memory specificity training (MEST) has also been used to combat memory biases in depression. In MEST, participants are trained to recall specific details of their autobiographical memories. One focus of MEST is on making negative memories more specific; this is done to make negative memory less overgeneral (e.g., "I'm a failure") and specific memory (e.g., "I did not receive the promotion at work"; Raes, Williams, & Hermans, 2009). Both PMET and MEST training are associated with multiple benefits, including more specific positive memories, fewer intrusive negative memories, improved emotion regulation, and positive changes in mood state (Arditte Hall et al., 2018; Lang, Moulds, & Holmes, 2009; Raes et al., 2009; Watkins, Baeyens, & Read, 2009).

Researchers have also started to investigate the possibility of training cognitive control in depression and of examining the effects of this training on emotion regulation. Joormann, Hertel, LeMoult, and Gotlib (2009), for example, showed that depressed participants can be trained to forget negative material when provided with a strategy for how to keep irrelevant material out of working memory. These training paradigms typically ask participants to practice removing irrelevant negative material from working memory and replacing it with relevant positive or neutral materials for later recall. Brief interventions that target increasing cognitive control in severely depressed outpatients have, in turn, led to significant decreases in both depressive symptoms and rumination (Siegle, Thompson, et al., 2007). Moreover, Jopling, Gotlib, and LeMoult (2020) documented that depressed participants who

completed cognitive control training also exhibited more adaptive neuroendocrine and physiological responses to stress compared to depressed participants who received sham training.

Thus, taken together, findings suggest that cognitive biases can be modified in relatively few training sessions using easily implemented designs. Modifying cognitive biases has, in turn, been linked to short-term improvements in mood and stress reactivity, which highlights the interconnection between cognitive and psychobiological responses to stress (LeMoult, 2020) and suggests that such training might play an important role in treatment. Future research is needed to extend these investigations to a clinical population, and to examine the persistence of cognitive changes and mood states over longer durations of time.

ER Strategies

Consistent with some of the studies on cognitive biases, researchers interested in ER have found that ruminating about negative thoughts, feelings, and events is linked with depression. In contrast, research findings suggest that reappraisal is an effective strategy to shift individuals' affect from negative to positive. These findings highlight a clear convergence between basic psychopathology research and applied psychotherapy research. One of the best examples of this convergence is a new therapy focused on reducing engagement in maladaptive emotion regulation strategies. Rumination-focused CBT for depression (Watkins, 2018) is an EST centered around helping clients to shift from ruminative thinking to more action-oriented, concrete, specific, and experientially focused thoughts and behaviors.

As mentioned earlier, helping clients to change their cognitive appraisals has been identified as a key principle of change underlying psychological treatments that work for depression. Similarly, the basic finding that emotional disclosure has a positive effect on mood and functioning, as opposed to the negative impact of emotion suppression, is also consistent with another principle of change identified earlier (i.e., improving awareness, acceptance, and regulation of emotions). Interestingly, however, while therapists from different theoretical orientations have consistently recognized the importance of changing clients' views of self and others, and employ a range of techniques to do so (e.g., reformulation, or cognitive restructuring), an explicit focus on the experience and processing of emotion is emphasized to a greater and lesser extent across ESTs. Brief psychodynamic, process–experiential and interpersonal therapies, exposure-based CBT, and newer emotion-focused CBT (e.g., Unified Protocol for Emotional Disorders; Barlow et al., 2010) tend to place the greatest emphasis on emotional processing in session. Moreover, emotional activation has been linked with treatment outcome across a variety of theoretical orientations (Peluso & Freund, 2018). Not surprisingly, this has been the case in experiential and humanistic therapies, such as emotion-focused therapy (EFT; Greenberg & Watson, 2006), which specifically and explicitly focus on the processing and transformation of emotion (see Elliott, Watson, Greenberg, Timulak, & Freire, 2013). However, emotional deepening has also been associated with treatment outcome in traditional CBT (e.g., Castonguay, Goldfried, Wiser, Raue, & Hayes, 1996; Watson & Bedard, 2006), an important finding given that cognitive behavioral therapist tend to try to control or reduce emotions—at least compared to psychodynamic therapy (Blagys &

Hilsenroth, 2000). Thus, both basic and psychotherapy research appear to suggest that therapists might foster clients' improvement by facilitating emotional disclosure and expression, irrespective of their preferred theoretical orientation.

While basic research findings on reappraisal and suppression are consistent with principles of change derived from psychotherapy research, this is not the case for distraction. Distraction has received only limited attention in EST treatments for depression despite evidence from basic research that it is an effective ER strategy that is less effortful, and therefore more accessible, to those who are severely depressed. One exception is dialectical behavioral therapy (Linehan, 1993), a CBT that includes a module on distress tolerance skills, in which clients are taught to use strategies to self-soothe and distract themselves during crises. Clinicians and scholars of different orientations are encouraged to complement their clinical repertoires with interventions that use distraction to help depressed clients to disengage, at appropriate times and places and for limited periods, from rumination and negative emotions. For example, one of the authors of this chapter (L. G. C.) has learned that blasting Beethoven's Ninth Symphony in response to pervasive and recurrent ruminations late at night is not incompatible with (and may actually be a powerful adjunct to) verbally oriented forms of therapy. In other words, although there are times to explore emotions and examine maladaptive cognitions, there are also times when individuals would benefit from taking their minds away from them, for example, at bedtime or when they need to be present and engaged in other interpersonal or occupational activities.

Stress

Research conducted within both biological and psychological domains of psychopathology has demonstrated that stressful life events have an impact on the onset and maintenance of depression. A central implication of such findings is that therapists who are working with depressed clients should be prepared to use interventions to help clients decrease their levels of stress and to deal more effectively with the stressors they encounter in their lives.

Most CBTs for depression include teaching relaxation, coping, and problem-solving skills to help patients manage ongoing stressors (see Follette & Greenberg, 2006). Of note, exposure-based cognitive therapy (EBCT)—a newer CBT for depression—makes stress management a primary focus in an attempt to build on and increase the efficacy of current ESTs for depression (see Hayes, Feldman, Beevers, Laurenceau, & Cardaciotto, 2007). The exposure-based phase of this therapy involves activating and emotionally processing key experiences and feelings of worthlessness, hopelessness, and defectiveness that drive individuals' depression. However, depressed individuals who seek treatment frequently experience high levels of distress and depletion of resources, and their depressive symptoms and attempts to cope with stressors can generate even more stress. These problems may make processing core beliefs challenging and counterproductive. Therefore, the initial stage of this treatment aims to increase resources that build resilience, motivation, and stability in preparation for exposure. This initial stabilization phase involves emotion regulation and mindfulness meditation techniques to help clients recognize and reduce avoidance and rumination, behavioral activation exercises, problem solving,

and strategies for increasing the health and regularity of sleep, eating, and exercise patterns. The purpose of this first phase of therapy is to teach stress management skills and to reduce maladaptive processes, including avoidance, rumination, and stress generating behaviors that maintain depression. These skills are thought to be critical both to prepare for deeper emotional and cognitive change and to prevent relapse.

Importantly, there is no reason to assume that the addition of these stress-regulating strategies should be restricted to CBT-based interventions. They are not likely to interfere with flexible and competent use of traditional interventions prescribed in any EST-based treatment for depression, and they may help individuals to better cope with stressors they encounter.

Interpersonal Factors

Research on interpersonal vulnerability for depression underscores the influence of adverse interpersonal relationships, especially early caregiving relationships, in the pathogenesis of depression. As such, more adaptive relational patterns may be core to achievement of successful therapy. Alexander and French (1946) posited that clients improve in psychodynamic therapy not because they acquire insights about previously unconscious conflicts, but because therapy provides them with a corrective emotional experience. Therapeutic change takes place, more precisely, when the therapist reacts to the client in a way that disconfirms the client's negative expectations of others that the client derived from the relationship with their parents or other early attachment figures. While the construct of corrective experience emerged from the psychodynamic literature, it is now recognized as a core mechanism of change that cuts across different orientations (Castonguay & Hill, 2012).

There can be no doubt that the relational stance embraced in all ESTs for depression (including CBT; see Castonguay, Youn, Xiao, & McAleavey, 2018) sharply contrasts with the neglect (e.g., inattentiveness, withdrawal, emotional unresponsiveness) and abuse (e.g., overt hostility, antipathy) that have been experienced by some depressed individuals. Interacting with clients in an attentive, warm, engaged, empathic, accepting, affirming, autonomy-granting, calm, structured, and consistent way (as promoted in the writing of leading clinicians of different orientations; e.g., Carl Rogers, Lorna Benjamin, Marsha Linehan, and Marvin Goldfried), is likely to differ from the way many depressed people have been treated. This new way of being treated violates the client's expectations that they will be treated poorly by a person with whom they have shared personal information and have been emotionally vulnerable, one whom they may view as an authority figure, and with whom they have developed an attachment, not unlike that with a parent. As described elsewhere (see Hill et al., 2012), this disconfirmation of expectations may facilitate the reduction of symptoms, change the view of self and others, and allow the client to test out, practice, and integrate more adaptive interpersonal patterns.

In contrast, interacting in a hostile and dismissive way might reenact old aversive interpersonal patterns and prevent change. This is consistent with a number of applied psychotherapy studies. For example, fine-grained analyses of therapist–client transactions indicated that poorer outcomes were associated with therapists using "blaming" or complex communications (i.e., seemingly "supportive" statements that

also convey hostility or criticism; Henry, Schacht, & Strupp, 1986). In a subsequent study, Henry, Schacht, and Strupp (1990) found that therapists exercised more disaffiliative interpersonal behaviors (e.g., belittling and blaming, ignoring and neglecting) with clients who had a poor outcome, as defined by no change in how they treated themselves, compared to those clients who had a good outcome. For those with a good outcome, therapists displayed an almost complete absence of such negative interpersonal processes. These authors also found a significant positive correlation between the frequency of therapist statements marked by disaffiliative control and hostility and the frequency of patient utterances marked by self-criticism and self-blame. Altogether, these findings suggest that the contrast between the early experiences of many depressed clients and the experience of a supportive relationship with a therapist may mediate (at least in part) the positive link between the working alliance and outcome across different forms of therapy.

In addition to underscoring the importance of adopting an interpersonally warm and involved attitude, research findings indicate that clients may benefit from exploring early relationship events, including traumas, and their ramifications relative to clients' relationships with others during their life. This is at the core of at least two ESTs: process–experiential (i.e., emotion-focused) and short-term psychodynamic therapies. Consistent with the emotion-focused model, the resolution of unfinished business with significant others has been linked with therapeutic change (Elliott et al., 2013). In line with psychodynamic theory, process research indicates that the therapist's accurate interpretation of core relationships in the client's life is linked with positive outcomes (Crits-Christoph & Connolly Gibbons, 2002). Interestingly, a focus on past relationships and the pattern of interactions they might have influenced may also be beneficial in other forms of treatment. For example, while process studies indicate that CBT focuses less on developmental and interpersonal issues than does psychodynamic therapy (see Blagys & Hilsenroth, 2000), other investigations suggest that it might be helpful when CBT therapists do focus on these issues. For example, Hayes, Castonguay, and Goldfried (1996) found that the degree to which CBT therapists focused on attachment with early caregivers predicted therapeutic outcomes at the end of treatment and relapse rates 2 years after therapy. In addition, a study by Jones and Pulos (1993) suggests that CBT therapists may improve treatment outcomes by eliciting and processing memories of interpersonal experiences using the following psychodynamic techniques: linking patients' feelings or perceptions to situations or behavior of the past; discussing memories or reconstructions of childhood; drawing connections between the therapeutic relationship and other relationships; and identifying recurrent themes in patients' experience or conduct.

These findings suggest that generating emotional memories in therapy for depression might be helpful, a finding that will not surprise practitioners of CBT for PTSD, exposure-based methods for depression, and psychodynamic therapies. However, given the presence of memory biases and ruminative tendencies in depression, the timing of interventions that elicit emotional memories need to be considered carefully in the context of the therapeutic relationship and the client's current functioning, resources, and ability to regulate their emotions. Moreover, these interventions are not necessary in all cases, and therapists should use their skills in case conceptualization and progress monitoring to determine whether and when these techniques are indicated.

In addition to the developmental influence of interpersonal events and relationships, basic findings also point to the importance of addressing current interpersonal issues. Phenomenological research on interpersonal irritability and marital discord provides support for some of the specific principles of change underlying ESTs for depression, in particular IPT. Moreover, research on interpersonal factors that are involved in the etiology of depression can inform more specific treatment targets. When working with depressed clients, therapists should inquire about and, if necessary, work to help change clients' tendency to seek excessive reassurance and verification of their negative view of self, as well as other aversive behaviors that generate interpersonal stress and subsequent depression. As discussed by Joiner and Timmons (2009), such foci of interventions are not explicit components of current IPT. While they are consistent with principles of change underlying behavior therapy (i.e., "decreasing reinforcement for depressive and avoidant behavior"; Follette & Greenberg, 2006, p. 94), full recognition of these issues may further improve the efficacy of traditional CBT for depression. For example, Hayes and colleagues (1996) found that relative to interpersonal issues, cognitive therapists tend to focus more on changing patients' cognitions about others than on direct interpersonal change. While the latter was linked with positive outcomes, the former was predictive of worse outcomes. Thus, attempting to foster concrete behavioral change in the interpersonal domain may be an important recommendation across many forms of therapy.

CONCLUSION

A great deal of research has been conducted on the psychopathology of depression. Studies investigating its phenomenology have revealed the preponderance, severity, recurrence, chronicity, and debilitating nature of this disorder. This descriptive research highlights several issues that therapists should assess and address when treating individuals with depression. Many of these issues are not frequently emphasized in the psychotherapy literature, including treatment implications of somatic symptoms, anger, poor health and occupational functioning, and timing of disorder onset.

Basic research on the etiology of depression has also identified several vulnerability factors involved in the onset and maintenance of this disorder. Some of these vulnerability factors are consistent with the principles of change that underlie currently validated psychological treatments for depression. However, other vulnerability factors have clinical implications that need to be more explicitly examined as ways to further improve evidence-based psychotherapy.

Basic research also conveys that the etiology of depression is not restricted to one set of causal or maintaining variables. Depression is likely to have multiple determinants. In addition, these determinants are interconnected (e.g., LeMoult, 2020), thereby creating complex networks of moderating and mediating influences on client's functioning. Accordingly, rather than restrict their understanding of depression within the confines of a single theory, clinicians are likely to increase their effectiveness by assessing an array of potentially underlying variables (biological, developmental, emotional, cognitive, behavioral, interpersonal). When forming a case conceptualization, clinicians are also encouraged to determine which of these

variables are likely to play primary roles for that specific client, and should therefore be core targets of their treatment.

In addition to targeting primary determinants of depression, the therapist's case formulation and treatment plan should take into account how these determinants interact with each other to interfere with the client's optimal level of functioning. While eclectic or prescriptive approaches to treating depression are likely to be effective (e.g., using social skills techniques with clients whose depression appears to be primarily related to their negative impact on others; using cognitive interventions when a highly functioning individual holds unreasonable standards), therapists should also develop integrative frameworks that assimilate different dimensions of functioning into broad and cohesive models (e.g., Gotlib & Hammen, 1992). Moreover, consistent with both prescriptive and integrative approaches, it is important to keep in mind that improvement in one domain of functioning is likely to have a positive impact on other domains. Adopting a more affiliative way of relating to others not only helps a client to find more fulfillment in relationships but is also likely to have an impact on their views of self. Therefore, therapists should assess multiple effects of interventions as part of ongoing outcome monitoring and in refining the treatment plan as therapy progresses. However, it is also important for the therapist to consider that a set of interventions may not have an impact on all potential determinants of depression; thus, they should broadly assess a client's functioning before terminating therapy, even if substantial changes have been achieved in one core dimension of functioning.

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CHAPTER 3

Generalized Anxiety Disorder

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Generalized anxiety disorder (GAD) has been conceptualized as the "basic" anxiety disorder given its persistent course and the features it shares with other anxiety disorders (Brown, Barlow, & Liebowitz, 1994). Accordingly, understanding the etiology and maintenance of GAD may have implications for understanding all anxiety disorders (Roemer, Orsillo, & Barlow, 2002).

TYPICAL SYMPTOMS AND ASSOCIATED FEATURES

Worry, a central feature of GAD, has been defined as "a chain of thoughts and images, negatively affect-laden and relatively uncontrollable . . . an attempt to engage in mental problem-solving on an issue whose outcome is uncertain but contains the possibility of one or more negative outcomes . . . [and] relate[d] closely to the fear process" (Borkovec, Robinson, Pruzinsky, & DePree, 1983, p. 10). The definition was later extended to include anxious apprehension for future negative events (Borkovec, Ray, & Stoeber, 1998). Although worry is a common phenomenon, it can become a persistent and disabling source of emotional discomfort in GAD. Persons with GAD worry not only about major life issues (e.g., health, finances, relationships) but also minor issues that others would not perceive as threatening (e.g., household chores). Pathological worry often generates additional distress due to its catastrophic thought content, and it deviates from normal worry due to its pervasiveness (e.g., Niles, Lebeau, Liao, Glenn, & Craske, 2012) and uncontrollability (e.g., Hallion & Ruscio, 2013).

Cognitive and physiological features of GAD elucidate the nature and function of worry. Worry is predominated by negative verbal thought activity, such that when people worry, they talk to themselves about negative events that may occur in the future (Borkovec et al., 1998). Because of its threat-related content, worry generates increases in subjective distress and sympathetic nervous system activity, while also reducing parasympathetic nervous system activity (e.g., Llera & Newman, 2010, 2014; Steinfurth, Alius, Wendt, & Hamm, 2017). By generating distress and somatic activation, worry reduces the likelihood of further reactivity associated with a fearful event (e.g., Llera & Newman, 2010, 2014). Thus, worry may account for the autonomic inflexibility characteristic of GAD (e.g., Hoehn-Saric, McLeod, Funderburk, & Kowalski, 2004) and serve as a strategy to avoid emotional processing of unexpected negative events (Newman & Llera, 2011). Accordingly, persons with GAD report that worry creates sustained levels of negative emotion, while also helping them avoid sudden increases in negative emotion and prepare for the worst (Borkovec & Roemer, 1995; Llera & Newman, 2017).

GAD and worry impact information processing in ways that interfere with learning from experience and contribute to the maintenance of anxious meanings attributed to various stimuli. People with GAD demonstrate a bias for attending to threat cues (Goodwin, Yiend, & Hirsch, 2017) and hold exaggerated beliefs about the probability of negative events (Butler & Mathews, 1983). However, given that there is no actual danger, as it only exists in their future-oriented thought, the threat cannot be physically avoided. As long as danger is perceived, individuals with GAD continue to search for ways to avoid it or prepare themselves to cope with it. Thus, worry is a mental response to perceived threat when no behavioral solution exists. It primarily functions as a maladaptive coping behavior, a means to prepare for the worst and avoid large shifts in negative emotions (Borkovec & Roemer, 1995; Newman & Llera, 2011).

Cognitive biases in GAD also extend to memory for positively and negatively valenced information. For example, compared to persons without anxiety disorders, persons with GAD have a bias toward greater memory for threat-related words (Coles, Turk, & Heimberg, 2007). Persons with GAD also show deficiencies in probabilistic learning, with a particular tendency to underestimate the probabilities of positive outcomes (LaFreniere & Newman, 2019). Memory biases and probabilistic learning deficits may account for the persistent apprehensive expectations characteristic of GAD. For example, when individuals with GAD were asked to track the outcomes of their worries, 91% of worries did not come true, and 30% of worries that did come true turned out better than expected (LaFreniere & Newman, 2020). Furthermore, recording a greater number of untrue worries predicted a reduction in worry over time (LaFreniere & Newman, 2020), suggesting that memory for positive and negative information may be involved in the maintenance of GAD.

GAD is also associated with significant distress, disability, and impairment in quality of life. The most detailed information on disability and quality-of-life effects of GAD comes from the Epidemiologic Catchment Area (ECA) community survey (Robins & Regier, 1991), one of the largest investigations of the prevalence of mental disorders, and from a multisite survey of psychiatric outpatients (Massion, Warshaw, & Keller, 1993). The overall picture for GAD is one of significant psychosocial impairment, with 38% of ECA subjects and 71% of outpatients characterizing their

emotional health as only fair to poor; 27 and 25%, respectively, were receiving disability payments, and only about 50% were working full-time. Of those who were working, 38% had missed at least 1 week of work in the past year due to their anxiety (Massion et al., 1993). In the ECA, GAD was associated with a threefold greater likelihood of working at a low occupational level, and a more than a twofold likelihood of earning less than $10,000 per year (Robins & Regier, 1991). Studies suggest that role and quality of life impairment in those with pure GAD (without comorbidity) is comparable to pure major depressive disorder (MDD; Hoffman, Dukes, & Wittchen, 2008; Wittchen, Carter, Pfister, Montgomery, & Kessler, 2000). Moreover, high rates of suicidality can be found in persons with GAD. In one longitudinal epidemiological study in the United States, young adults (ages 16–25 years) diagnosed with GAD had a sixfold increased likelihood of suicidal ideation and a more than twofold greater likelihood of attempted suicide, after researchers controlled for other mental disorders and stressful life events (Boden, Fergusson, & Horwood, 2007).

GAD also has consequences for physical health, with 35% of individuals with GAD in the ECA sample and 23% of those in the Massion and colleagues (1993) sample rating their physical health as only fair to poor. GAD is associated with a number of comorbid medical problems such as asthma (Goodwin, Jacobi, & Thefeld, 2003), chronic pain conditions (Von Korff et al., 2005), irritable bowel syndrome (Gros, Antony, McCabe, & Swinson, 2009), and insomnia (Brenes et al., 2009). Furthermore, GAD is a risk factor for cardiovascular disease independent of depression (e.g., Batelaan, ten Have, van Balkom, Tuithof, & de Graaf, 2014; Martens et al., 2010). Persons with GAD are among the heaviest users of primary care, specialty clinic, and emergency room resources, contributing substantially to the nonpsychiatric cost associated with anxiety disorders in the United States (e.g., Fogarty, Sharma, Chetty, & Culpepper, 2008; Jones, Ames, Jeffries, Scarinci, & Brantley, 2001). Even individuals who do not meet the full criteria for GAD but have a subthreshold level of symptoms experience significant impairment, as well as greater health care use compared to individuals without GAD symptoms (Haller, Cramer, Lauche, Gass, & Dobos, 2014).

Finally, GAD has been associated with interpersonal dysfunction (Crits-Christoph, Gibbons, Narducci, Schamberger, & Gallop, 2005; Newman & Erickson, 2010; Przeworski et al., 2011). Individuals with GAD tend to report a predominance of worrisome thoughts related to relationship concerns (Breitholtz, Johansson, & Öst, 1999; Roemer, Molina, & Borkovec, 1997). They also exhibit a tendency to perceive social threats in their environment (Goodwin et al., 2017). In a like manner, GAD is more associated with marital discord or dissatisfaction than other anxiety or mood disorders (Whisman, 1999). Compared to those without GAD, individuals with GAD are more likely to have enmeshed relationships or to engage in role reversal, such as a child assuming parental responsibility (Cassidy, 1995; Cassidy, Lichtenstein-Phelps, Sibrava, Thomas, & Borkovec, 2009). People with GAD have disrupted perceptions of their negative interpersonal impact on others (Erickson & Newman, 2007), which can impact their relationships and maintain interpersonal dysfunction.

Treatment Implications

Individuals with GAD anxiously anticipate and engage in mental attempts to problem-solve and prepare for possible negative events. They frequently detect and interpret threat in their environment, even when no threat exists (Goodwin et al.,

2017). Their mental experience is dominated by worry, which is elicited by internal or external cues that signal real or imagined danger. Individuals with GAD use worry to figure out how to prevent bad things from happening or to prepare for the worst (Borkovec & Roemer, 1995). They spend a significant amount of time preoccupied with their worrisome thoughts and therefore tend to describe themselves as having been worriers all their lives. Chronic worry becomes a way of life to the point that these individuals may not realize there are treatments available to them. Since many people with GAD might not even seek treatment, dissemination of information about the helpfulness and availability of psychotherapy for GAD is an important first step.

Individuals with GAD may be so preoccupied with worries that they neglect the world around them. Generally anticipating the worst, they tend to absorb information in accordance with their negative expectations. They do not utilize information available to them via their emotional experience, which can adaptively motivate them, guide their behavior, or facilitate learning and subsequent growth (Borkovec, 2002). In their efforts to focus on imagined threat, they are distracted from the present, including any information that may be useful and/or make life less stressful. Despite its negative impact, GAD is unique compared to other anxiety disorders, in that worry is inherently reinforcing. Some individuals with GAD hold the superstitious belief that their worry can actually prevent negative things from happening. The relief they experience when the feared event does not occur reinforces their worry. They are firm in their conviction that if they had not engaged in worrisome thinking, the aversive event might have taken place. Other individuals with GAD believe that their motivation hinges on their worry. Without it, they would no longer be goal-oriented. They feel that they need to be in a constant state of activation, anticipating future catastrophes so they can avoid them.

Based on empirical findings on the symptoms and clinical features of GAD reviewed earlier, we can make a number of recommendations about how treatment might best proceed with such patients. Because worry is negatively reinforced, individuals with GAD need to gain insight that their worry is ultimately maladaptive. Although it is likely to take considerable therapeutic effort for patients with GAD to learn that their worry is not helpful, clinicians should assist them in developing a new perspective. As many individuals with GAD tend to associate their worry with a motivation to succeed, they need to learn to differentiate between emotional distress and drive and ambition. It is important for them to realize that they can still be goal-oriented in the absence of anxiety. The majority of their experience centers on trying to prevent future disasters; thus, their actions are perceived to be externally motivated. Accordingly, treatment should focus on helping them rediscover the intrinsic motivation behind their actions and come to understand that it is possible to engage in something even when not actively fearing negative consequences. Patients with GAD do not fully experience life because they are consumed by mentally creating, and subsequently avoiding, new dangers and threats. To maximize their experience, they need to refrain from attending to external consequences to the extent that they are disengaged from meaningful and rewarding activities. They need to recognize the negative impact their worry has on their well-being. Thus, before beginning to work with clients on decreasing worry, therapists need to help clients see that their worry is neither helpful nor necessary.

Given the potency of worry, it is also important to target and track this cognitive process consistently and not to treat GAD as simply diffuse anxiety. Therefore,

self-monitoring is likely to be a beneficial component of therapy, irrespective of the clinician's theoretical orientation. Clinicians need to help patients identify experiences that are inconsistent with their expectations. These patients are constantly attuned to imagined negative events that may happen in the future, so it is necessary for them to return to the present moment. Being present enables accurate tracking of information that can disabuse patients of maladaptive beliefs. When focused on what may happen in the future, people with GAD tend to lose information available to them that they can use to their benefit in their present environment. Therefore, it is important that clinicians help them actively make an effort to track and observe aspects of the world around them that are inconsistent with their apprehensions. Moreover, individuals with GAD need more adaptive ways to perceive, interpret, and predict events. They tend to reject information that contradicts their negative cognitions, often regarding positive experiences as flukes. Therefore, therapists need to teach clients to change negative beliefs. Overall, a major goal of therapy is to help these patients shift from inaccurate negative expectations to beliefs that are less anxiety provoking.

In addition to teaching new ways of seeing the world, more recently developed treatments have targeted the predilection of individuals with GAD to avoid emotional processing via worry. Given that emotional processing avoidance may be maintaining the generalized anxiety symptoms, active engagement in emotional processing may be an important component of treatment, particularly the processing of fear (Llera & Newman, 2010; Newman, Castonguay, Borkovec, Fisher, & Nordberg, 2008). However, given that worry precludes such processing, and because worry increases and maintains anxiety and stress, an important therapeutic task is to engage clients in the use of applied relaxation. In a relaxed state, emotional processing is more likely. Within therapy, exposing clients to sharp shifts in anxiety by having them relax and then imagine a worrisome outcome might be helpful. Outside of therapy, it is necessary for clients to expose themselves to sharp shifts in negative emotions, especially those occurring in worry-related situations, such as interpersonal relationships (Newman, Llera, Erickson, & Przeworski, 2014). Care should be taken to ensure that clients attempt to lower their baseline state of anxiety and tension during normal activity to one that mimics euthymia and naturally allow any mood shifts that may occur both during and outside of therapy.

Because GAD is associated with impairment in health and occupational functioning, it is recommended that clinicians subscribing to all theoretical orientations assess associated disability at different phases of treatment in order to determine ways in which clients can improve those important aspects of functioning. Similarly, therapists should assess for suicidal ideation, intent, and plan throughout the course of treatment, especially if it has been determined that the patient is at risk for suicide. Moreover, some of the somatic symptoms associated with GAD (e.g., fatigue, sleep disturbance, irritability, difficulties with concentration) can be addressed with specific interventions, such as relaxation and sleep hygiene techniques, and should be incorporated into treatment.

Also, because interpersonal problems are highly associated with GAD, therapists should assess interpersonal difficulties at different phases of therapy, including the end of the treatment. Remaining interpersonal problems at the end of therapy have been linked to poorer outcome at follow-up. Thus, addressing such problems before the end of treatment may prevent relapse. Both theory and research evidence

have suggested that interpersonal problems (past and current) are linked to the cause and/or maintenance of GAD. As such, interpersonal difficulties are further addressed in the section on etiology.

The aforementioned recommendations are beneficial in targeting the core cognitive, emotional, and behavioral symptoms of individuals diagnosed with GAD. However, individuals who fall below threshold may still be significantly affected by their GAD symptoms. Although their overall dysfunction may not be as severe as that of individuals with full GAD, their symptoms can be just as debilitating during periods of excessive worry and high anxiety. Therefore, it is important to assess clinically for the presence of subthreshold levels of GAD. Whereas a patient is generally treated within an overall diagnostic and therapeutic framework, including consideration of personality disorders, it may be necessary to consider the patient's needs from an individual standpoint. This is especially true if they do not meet criteria for a particular disorder but report some associated symptoms requiring stabilization.

COURSE AND OUTCOME

GAD has a common onset during middle age (median = 31–39 years; Kessler et al., 2005; Ruscio et al., 2017) and assumes a chronic course. In the Harvard–Brown Anxiety Research Program (HARP), a naturalistic prospective study of psychiatric patients, the probability of recovery from GAD over a 12-year period was 58%, and the probability of subsequent recurrence among those who did recover was 45% (Bruce et al., 2005). A separate prospective study of primary care patients with GAD revealed a 39% probability of full recovery after 2 years, as well as a 52% probability of recurrence of moderate or greater symptoms among recovered patients (Rodriguez et al., 2006). Thus, full recovery from GAD is rare and often short-lived. Factors associated with a longer course of GAD in the United States include being female and being of African American or Latinx ethnicity (Bjornsson et al., 2014; Rodriguez et al., 2006; Sibrava et al., 2013). Internationally, GAD takes on a more prolonged course in lower-income countries and among individuals with lower socioeconomic status (Ruscio et al., 2017).

Individuals with GAD experience varying levels of symptom severity and impairment over time. Among HARP participants with GAD, disorder severity declined modestly over 14 years, though participants did not improve beyond partial recovery on average (Ramsawh, Raffa, Edelen, Rende, & Keller, 2009). Just as some persons with GAD experience decreases in severity, others experience increases in severity, rising from subthreshold to threshold levels (Angst, Gamma, Baldwin, Ajdacic-Gross, & Rossler, 2009). Thus, while experiencing a chronic course, the typical person with GAD experiences fluctuations in symptom severity (e.g., Wittchen, Lieb, Pfister, & Schuster, 2000). As such, there may be periods when the severity of GAD symptoms is below threshold levels, without necessarily indicating recovery.

Cognitive-behavioral therapy (CBT) is currently the "gold standard" treatment for GAD. CBT for GAD produces significant improvement that is maintained for up to 2 years following treatment termination (Borkovec & Ruscio, 2001; Cuijpers et al., 2014). CBT has also been found to generate greater improvement than no treatment, psychodynamic therapy, pharmacotherapy, nondirective therapy, and placebo therapy (Borkovec & Ruscio, 2001; Cuijpers et al., 2014; Salzer, Winkelbach, Leweke,

Leibing, & Leichsenring, 2011). CBT is well liked by clients and associated with a lower dropout rate compared to pharmacotherapy (Mitte, 2005). Though findings are mixed, the literature generally demonstrates superior efficacy of CBT compared to either individual cognitive or behavioral therapy components alone (Borkovec & Ruscio, 2001; Cuijpers et al., 2014). Moderator analyses suggest that individual component treatments may be more effective for GAD clients with a longer course of the disorder (Newman & Fisher, 2013; Newman, Shin, & Lanza, 2019). These findings suggest that, while full CBT packages provide a range of efficacious therapeutic techniques, prolonged and intensive focus on individual techniques may help clients break deeply entrenched habits.

In addition to showing *statistically* significant improvement, CBT has also demonstrated *clinically* significant improvement. CBT reduces worry to normative levels and leads to recovery in a majority of clients, and these gains are typically maintained for up to 1 year (Covin, Ouimet, Seeds, & Dozois, 2008; Hanrahan, Field, Jones, & Davey, 2013). CBT interventions include self-monitoring, stimulus control, relaxation, self-control desensitization, and cognitive therapy. As described by Woody and Ollendick (2006) and Newman, Stiles, Janeck, and Woody (2006), these techniques reflect five principles of change (i.e., challenge misconceptions, actively test the validity of erroneous beliefs, use repeated exposure, eliminate avoidance, and improve skills) that can be implemented by interventions espoused in various approaches.

Benzodiazepines (BZs), buspirone, and tricyclic antidepressants were the first classes of medications to be approved by the U.S. Food and Drug Administration (FDA) for the treatment of GAD. However, BZs carry risk of sedation, motor and cognitive abnormalities, as well as physical dependency and withdrawal problems (e.g., van Laar, Volkerts, & Verbaten, 2001; Wang, Bohn, Glynn, Mogun, & Avorn, 2001). Because of their adverse effects, BZs have been relegated by most psychiatrists to treat only acute anxiety symptoms (Baldwin, Allgulander, Bandelow, Ferre, & Pallanti, 2012). In clinical trials, buspirone (Rickels et al., 1982; Volz, Möller, & Sturm, 1994) and the tricyclic antidepressant imipramine (Kahn et al., 1986; Rickels, Downing, Schweizer, & Hassman, 1993) have shown efficacy.

Since 1999, a number of studies have documented the efficacy of selective serotonin reuptake inhibitor (SSRI) and serotonin–norepinephrine reuptake inhibitor (SNRI) antidepressants in the treatment of GAD. Venlafaxine XR (extended release) was the first of these drugs to be approved by the FDA for the indication of GAD. In large-scale studies, venlafaxine XR demonstrated significantly better clinical efficacy than placebo after 8 weeks (Rickels, Pollack, Sheehan, & Haskins, 2000; Sheehan, 1999), and after 6 months (Allgulander, Hackett, & Salinas, 2001; Gelenberg et al., 2000) of treatment. Paroxetine was also studied in several GAD trials (Davidson, 2001; McCafferty, Bellew, Zanelli, Iyengar, & Hewett, 2000; Pollack et al., 2001; Rickels et al., 2003; Stocchi et al., 2003) and approved by the FDA for the treatment of GAD. Additional SSRI/SNRI drugs approved by the FDA for the treatment of GAD include duloxetine and escitalopram.

Despite the efficacy of SNRIs and SSRIs in the treatment of GAD, low remission rates have been observed during the acute phase of treatment of GAD for venlafaxine XR (37%; Gelenberg et al., 2000) and paroxetine (33–39%; Rickels, Rynn, Iyengar, & Duff, 2006). Although remission rates are higher after longer-term treatment with venlafaxine XR (79%; Rickels et al., 2013) and paroxetine (73%; Stocchi et al., 2003),

it is equally important to attempt to increase these rates during acute treatment in order to improve compliance and relieve suffering and functional impairment as quickly as possible. Thus, it is important to test new ways of improving short-term pharmacological treatment of GAD.

Treatment Implications

Given GAD's chronic fluctuating course, it is unlikely that GAD symptoms and associated features will permanently resolve themselves over time in the absence of treatment. Although symptoms may decrease unaided, they can resurface to the same degree, especially in response to stressors. In patients seeking treatment, GAD symptoms may appear to have been successfully treated when assessed at discrete time points. Likewise, patients presenting with subthreshold GAD may be at risk to develop full-blown GAD. Psychotherapists should use standards employed in epidemiological studies to track GAD symptoms and better understand their ebb and flow, such as the symptom course indicative of sustained recovery. Also, it is recommended that treatment not be discontinued until clients have shown sustained recovery.

Although data suggest that GAD is chronic, clinicians should also be aware that patients may inaccurately report their symptoms. For instance, if they are feeling anxious in a given moment or for an extended period, they are more likely to report that they have been that way for the past 6 months, thus meeting the chronicity criterion for GAD. Therefore, it is important for therapists to help patients increase their awareness of their patterns of functioning. Furthermore, individuals with GAD rarely experience full or sustained recovery, so it is imperative that therapists approach treatment with realistic expectations and aim not for a "cure" but to help the client to learn how to cope with the disorder. The chronicity and recurrence of GAD also suggests that clients may need long-term therapy. Therapists should incorporate methods to predict and prevent relapses, as well as cope with possible resurgences of the disorder. Acceptance-based therapy techniques, such as mindfulness, may also help clients to stop bracing for the worst and prevent full-blown relapses (Whiteside et al., 2007).

EPIDEMIOLOGY

Variations in definition, including subsequent changes to DSM criteria, have interfered with precise prevalence estimates for GAD. However, several epidemiological studies employing large representative samples have approximated actual prevalence rates. Lifetime prevalence of DSM-III-R GAD, excluding comorbid panic and depression, was 3.6% (Kendler, Neale, Kessler, Heath, & Eaves, 1992). The National Comorbidity Survey found a 12-month prevalence of 3.1% and a lifetime prevalence of 5.1% (Wittchen, Zhao, Kessler, & Eaton, 1994). The National Comorbidity Survey Replication study indicated a lifetime prevalence for DSM-IV GAD of 5.7% and a 12-month prevalence of 3.1% (Kessler et al., 2005). Projected lifetime prevalence at age 75 is 8.3%. Compared with DSM-IV GAD prevalence rates, the World Mental Health Surveys report a higher 12-month prevalence of DSM-5 GAD of 4.0% and a lifetime prevalence of 7.8% for the United States (Ruscio et al., 2017). Globally, a lifetime

DSM-5 prevalence of 3.6% and a 12-month prevalence of 1.8% has been found in the 26 participating countries (Ruscio et al., 2017).

The presence of GAD is associated with a number of sociodemographic features. Gender is one of the factors most consistently associated with GAD, with rates of the disorder approximately two times greater in women than in men in community (Wittchen et al., 1994) and clinical (e.g., Woodman, Noyes, Black, Schlosser, & Yagla, 1999) samples. Furthermore elevated rates of GAD have been found in previously married (Wittchen et al., 1994) and never married (Ruscio et al., 2017) individuals compared to married individuals. Homemakers, individuals not working outside the home, and unemployed individuals also have elevated rates of GAD (Ruscio et al., 2017; Wittchen et al., 1994). Within the United States, prevalence is greatest in the Northeastern region (Wittchen et al., 1994), and globally, prevalence is greatest in higher income countries (Ruscio et al., 2017). Prevalence increases across adulthood, beginning at 4.1% for ages 18–29, 6.8% for ages 30–44, rising to 7.7% for ages 45–59, before decreasing to 3.6% for ages 60 and older (Kessler et al., 2005). Epidemiological surveys in the United States have found few differences in the prevalence of GAD as a function of ethnicity, religion, income, education and urbanicity (Wittchen et al., 1994).

Treatment Implications

GAD prevalence rates depend on the current diagnostic criteria. However, it is not uncommon for anxious patients to worry excessively or meet criteria for subthreshold GAD. Therefore, therapists should understand the determinants of GAD and related symptoms, since they are likely to encounter worrisome patients. Although patients may not meet full criteria for GAD, clinicians of all experience levels should not neglect subthreshold GAD symptoms because the emergence of full GAD may be prevented through early detection of symptoms. Given a patient's sociodemographic features, therapists should be aware that differential risk of GAD may be associated with different determinant stressors, such as divorce or the death of a loved one. Whereas ethnicity, education, religion, and income are not related to increased incidence of GAD, the content of patients' worries may differ depending on their sociodemographic background. Thus, clinicians need to understand the manifestations of GAD within the context of an individual's culture, ethnicity, religion, education, and income level.

COMORBIDITY

GAD is frequently comorbid with other psychological disorders. Current comorbidity for GAD has been reported to range from 6 to 29% for dysthymia, 39 to 76% for major depression, 23 to 44% for social phobia, 12 to 44% for panic disorder, and 12 to 25% for specific phobia—the five most common comorbid diagnoses (Brown, Campbell, Lehman, Grisham, & Mancill, 2001; Lamers et al., 2011; Ruscio et al., 2017; Wittchen et al., 1994). Reported rates of lifetime comorbidity for GAD range from 53 to 86% for major depression and 52 to 74% for other anxiety disorders (Brown et al., 2001; Judd et al., 1998; Lamers et al., 2011; Ruscio et al., 2017). High levels of lifetime

comorbidity have also been reported between GAD and substance used disorders (22%) and personality disorders (38%) (Dyck et al., 2001; Ruscio et al., 2017).

Comorbidity occurs both concurrently and sequentially in GAD (Moffitt et al., 2007). GAD has been found both to precede and to follow substance use disorders, panic disorder, and major depression (Grant et al., 2009; Moffitt et al., 2007; Wittchen, Kessler, Pfister, & Lieb, 2000). Moreover, the presence of GAD at one time point increases the probability of developing a different disorder at a later time point. In the National Comorbidity Survey Replication, current GAD significantly predicted subsequent panic disorder, posttraumatic stress disorder, mood disorders, and substance use disorders; even subthreshold GAD significantly predicted new onsets of panic disorder, mood disorders, and substance use disorders (Ruscio et al., 2007).

Because GAD is highly comorbid with other disorders, particularly MDD, some researchers have questioned whether GAD is a distinct disorder. However, considerable evidence supports the distinctiveness of GAD. Pure GAD and MDD produce comparable impairment, and they produce further impairment when comorbid (Hoffman et al., 2008). Moreover, the relationships among GAD symptoms are stronger than their relationships with MDD symptoms, and these relationships are stable over time (Beard et al., 2016). Beyond differences in clinical presentation, GAD and MDD have distinct risk factors (Beesdo, Pine, Lieb, & Wittchen, 2010; Kendler, Gardner, Gatz, & Pedersen, 2007) and neurobiological features (Oathes, Patenaude, Schatzberg, & Etkin, 2015; Weinberg, Kotov, & Proudfit, 2015). GAD and MDD are also equally likely to be temporally primary (Moffitt et al., 2007), refuting the notion that GAD is a mere prodrome for MDD. Thus, the literature supports the validity of GAD as a distinct disorder.

Comorbidity is associated with further impairment. Compared to those with pure GAD, persons with comorbidity in addition to GAD report more disability days (Hunt, Issakidis, & Andrews, 2002), interpersonal problems (Judd et al., 1998), and lower life satisfaction (Henning, Turk, Mennin, Fresco, & Heimberg, 2007). Moreover, compared to those with pure GAD, those with comorbidity generate greater health care costs due to medication use, hospitalization, and outpatient care (Sandelin, Kowalski, Ahnemark, & Allgulander, 2013). Having comorbid disorders in addition to GAD predicts a more prolonged course of both GAD (Bruce et al., 2005; Bruce, Machan, Dyck, & Keller, 2001; Kessler et al., 2002) and the comorbid disorders (Bruce et al., 2005; Kessler et al., 1996, 1998). Finally, comorbidity in GAD is associated with greater risk of suicide and all-cause mortality (Boden et al., 2007; Phillips et al., 2009).

Comorbid interpersonal factors such as marital tension (Durham, Allan, & Hackett, 1997) and personality disorders (Skodol, Geier, Grant, & Hasin, 2014) predict a more prolonged course of GAD, including poorer treatment outcomes. Borkovec, Newman, Pincus, and Lytle (2002) found that CBT for GAD failed to make a significant change in most interpersonal problem domains. Clients endorsing pretherapy interpersonal problems associated with dominance (e.g., domineering/controlling, intrusive/needy, vindictive/self-centered) responded least favorably to CBT. Tailoring CBT to address interpersonal problems may improve treatment outcomes for some clients. For example, behavior therapy outperformed both CBT and cognitive therapy in the treatment of clients with problems related to dominance (Newman, Jacobson, Erickson, & Fisher, 2017), and the addition of interpersonal and emotion processing techniques to traditional CBT led to better treatment outcomes among

clients with dismissive attachment styles (Newman, Castonguay, Jacobson, & Moore, 2015). Identifying and addressing clients' interpersonal problems can maximize the effectiveness of treatment for GAD.

Treatment Implications

GAD is a highly comorbid disorder that is associated with significant impairment whether or not it is the primary disorder. Clinicians are advised to periodically assess for the presence of other disorders when treating GAD, in order to adjust treatment in accordance with the client's difficulties. Assessing psychopathology at the end of treatment also allows clinicians to determine whether therapy alleviated comorbidity. The detection of residual comorbid problems might call for additional targeted interventions, particularly for interpersonal problems, because these tend to predict poor outcome after the completion of therapy. Similarly, given the high level of comorbid GAD observed in other disorders, especially depression, clinicians are advised to assess for GAD at different phases of treatment, even when it is not the primary disorder.

Conceptualization and treatment planning may differ depending on comorbidity, such that one disorder or related symptoms can affect the expression of another disorder. For example, a diagnosis of panic disorder alongside GAD can be associated with increased behavioral avoidance and may require situational exposure techniques. GAD can also be associated with symptoms of other disorders in the absence of those disorders. For instance, individuals with GAD may engage in behavioral avoidance of worry triggers. Thus, a thorough assessment of symptoms of associated disorder symptoms, including avoidance, is important. Addressing comorbid problems in patients with GAD may be critical for long-term recovery from GAD.

ETIOLOGY

The development of GAD is caused by the interplay of genetic, biological, environmental, and psychological factors. According to twin studies (e.g., Kendler et al., 1992), GAD is moderately regulated by genetic factors, such that about 30% of the variance is due to heritability. Similarly, heritable traits related to temperament, such as neuroticism, are strongly associated with anxiety disorders (Hettema, Neale, Myers, Prescott, & Kendler, 2006). Although genetic influences alone do not cause GAD, heritability contributes to a general biological predisposition to anxiety, which interacts with other elements in the emergence of a distinct disorder (e.g., Beesdo et al., 2010). A genetic vulnerability to anxiety manifests as GAD when the necessary biological, psychological, and environmental elements converge.

Learning theories illuminate factors in the development and maintenance of anxiety, and therefore suggest possible pathways in the etiology of GAD. Classical conditioning theory suggests that a child comes to fear a conditioned stimulus (CS) following repeated pairings with an aversive unconditioned stimulus. The CS generates anxiety because the child expects it to result in an aversive outcome. Anxious behavior is reinforced and maintained upon avoiding the CS (Mowrer, 1939). Operant conditioning also contributes to the development and maintenance of anxiety

when some parents foster children's anxious behavior by providing attention, reassurance, and assistance with avoidant behavior (Ayllon, Smith, & Rogers, 1970). Learning from social modeling may also contribute to anxious behavior (Bandura & Menlove, 1968).

Early attachment likely plays a role in the development and maintenance of GAD. Bowlby (1982) suggested that when a child's attachment figure is in some way absent, the child fails to develop a secure base from which to explore their environment. Subsequently, the child may view the world as unpredictable, uncontrollable, and threatening, thereby exaggerating the perceived likelihood and negative impact of feared events, and undermining their perceived ability to cope with these events. In support of Bowlby's model, adults with GAD have been found to report elevated rates of both anxious–ambivalent and avoidant attachment to their parents, compared to nondisordered control participants (Newman, Shin, & Zuellig, 2016).

It is possible that cognitive biases characteristic of GAD develop prior to the onset of GAD and serve as risk factors for the disorder. Although there are currently no empirical data to support this, a child might learn that bad things happen unexpectedly and develop the maladaptive coping strategy of repeatedly interpreting ambiguous information as an indication that something bad has happened or will happen, as a means to prepare for the worst. Models of GAD implicate processes such as worry about worry (Wells, 2005), intolerance of uncertainty (Dugas, Buhr, & Ladouceur, 2004), and fear of a negative emotional contrast (i.e., the fear of a sharp shift in negative emotion or being emotionally caught off guard) as possible cognitive factors that precede the development of GAD (Newman & Llera, 2011).

Children with GAD tend to have apprehensive expectations for the future, hold a negative self-image, and seek constant reassurance (Masi et al., 2004). These children also tend to have fewer friends and are less assertive than nonanxious children (Scharfstein, Alfano, Beidel, & Wong, 2011; Scharfstein & Beidel, 2015). Anxious children's fear and negative expectations of social situations may result in negative interpersonal cycles (Safran & Segal, 1990) in which these children influence others' behavior in a manner that reinforces their negative expectations and contributes to the maintenance of their anxiety. Indeed, children displaying anxious behavior tend to be judged less favorably by peers (Baker, Hudson, & Taylor, 2014), and adolescents' GAD symptoms predict greater levels of maternal criticism in subsequent years (Nelemans, Hale, Branje, Hawk, & Meeus, 2014). Thus, interpersonal dysfunction may be a pathway by which GAD develops and is maintained.

During childhood, individuals with GAD may learn that caring for others elicits love, approval, and nurturance. Extending this pattern into adulthood, they are frequently concerned about interpersonal relationships (Breitholtz et al., 1999; Roemer et al., 1997). Thus, worry may function as a means of anticipating the needs of, and threats to, significant others in the pursuit of satisfying interpersonal needs. Accordingly, persons with GAD display an attentional bias to social threat information (Goodwin et al., 2017) and are especially attuned to others' negative mental states when worried (Zainal & Newman, 2018). At the same time, persons with GAD and elevated worry are inaccurate in perceiving the impact of their social behavior on others (Erickson et al., 2016; Erickson & Newman, 2007; Shin & Newman, 2019). Considering these cognitive tendencies in concert, it follows that individuals with GAD would show not only greater interpersonal distress but also greater rigidity in their

interpersonal styles across differing situations than nonanxious people (Przeworski et al., 2011). Histories of role-reversed and enmeshed relationships may also cause persons with GAD to experience unresolved anger toward their primary caregivers (Cassidy, 1995; Cassidy et al., 2009), exacerbating their interpersonal difficulties.

Treatment Implications

Although many biological, psychological, and social factors are likely to be involved in the development of GAD, it is important to keep in mind that the etiology of a disorder does not necessarily have a bearing on its maintenance. Whereas people come to behave in certain ways as a result of some etiological factor, the behavior often persists in the absence of its origin. Often it is difficult to determine how a disorder develops, but recovery is possible if maintaining factors are identified and addressed. Therapists should investigate factors that may contribute to their GAD patients' worrisome and anxious states, especially since these patients tend to engage in behaviors that are perceived to be helpful but ultimately may exacerbate their problem. Additionally, it is important to monitor for secondary gains that enable patients' conditions. For example, GAD is often associated with reassurance seeking. Individuals with this disorder work themselves into an excessively worried state, then want validation from significant others that they are OK. They learn to associate worry with reassurance, which promotes further worry.

Once maintaining factors have been identified, therapy can center on eliminating them. Many of these factors have been targeted in CBT protocols. We should mention, however, that CBT techniques such as relaxation, stimulus control, social skills training, and exposure can be incorporated into treatment, irrespective of the therapist's theoretical approach. As we mentioned earlier, these interventions can be viewed as manifestations of general principles of change that could be implemented by techniques typically prescribed in other orientations (e.g., cognitive distortions and processes can be addressed by the use of confrontation and interpretation, as well as by cognitive restructuring; see Castonguay, 2000; Newman et al., 2006). Similarly, treatments for GAD have involved interpersonally focused psychotherapies to address interpersonal problems in GAD (Crits-Christoph et al., 2005; Newman et al., 2008, 2011), underscoring the importance of integrating therapy techniques in the treatment of GAD.

CONCLUSION

Worry is a universal experience that enables people to deal with challenging future events. Since this thought style is more abstract and less concrete (less image producing) than other types of thinking (Borkovec & Inz, 1990), it allows for abstract processing of emotional information, thereby facilitating avoidance of reactivity and surprising negative events. Thus, it functions as a strategy to manage the unpredictability and uncontrollability of potentially threatening future events. Nevertheless, the worry experienced by individuals with GAD is a pervasive, uncontrollable daily activity that often elicits emotional discomfort. Although patients with GAD engage in worry as a means to combat potential threat, it serves to exacerbate their

anxiety. Accordingly, GAD can be associated with debilitating distress, disability, and impairment in quality of life.

Overall, GAD is a highly chronic and comorbid anxiety disorder whose etiology is linked to a combination of genetic, biological, environmental, and psychological factors. Its core and associated symptoms are the products of the interaction among cognitive, imaginal, and physiological responses to continuously perceived threat (Borkovec & Inz, 1990; Newman & Borkovec, 2002). Although GAD is the least successfully treated anxiety disorder given its chronicity and pervasiveness (Brown et al., 1994), CBT is an efficacious treatment for GAD. CBT targets each of the aforementioned response systems, and interventions include self-monitoring, stimulus control, relaxation, self-control desensitization, and cognitive therapy. Although DSM-5 brought significant changes to the diagnostic criteria for anxiety disorders, worry continues to play a central, prominent role in GAD's conceptualization (American Psychiatric Association, 2013). Therefore, when treating GAD, clinicians and therapists should assist patients in recognizing the detrimental impact of their worry on their psychological and emotional functioning, and identifying and tracking their worrisome thinking, as well as underscore those experiences and aspects in the patients' environment that are inconsistent with their schemas to remediate maladaptive thought and behavior patterns.

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